
Alma Mater Studiorum – University of Bologna
Department of Computer Science and Engineering

**Wireless Systems (1) –
Physical Systems, Signals and Wireless Systems' Design**



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Figure-credits: some figures have been taken from slides published on the Web, by the following authors (in alphabetical order):

J.J. Garcia Luna Aceves (ucsc), James F. Kurose & Keith W. Ross, Jochen Schiller (fub), Nitin Vaidya (uiuc)



Background on wireless PHY layer

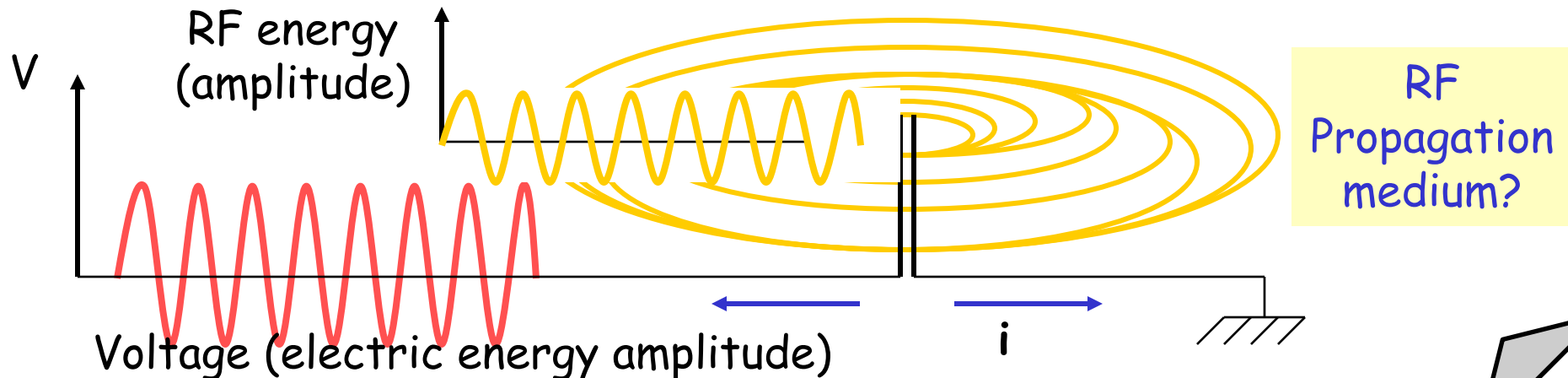
RF Properties

■ Understanding Radio Frequency

- Generation, coverage and propagation issues
- Fundamental for wireless planning and management
- <http://faraday.physics.utoronto.ca/PVB/Harrison/Flash/EM/EMWave/EMWave.html>
- http://wwwhome.ewi.utwente.nl/~ptdeboer/ham/xnecview/dipole_anim.html

■ Radio Frequency Signals

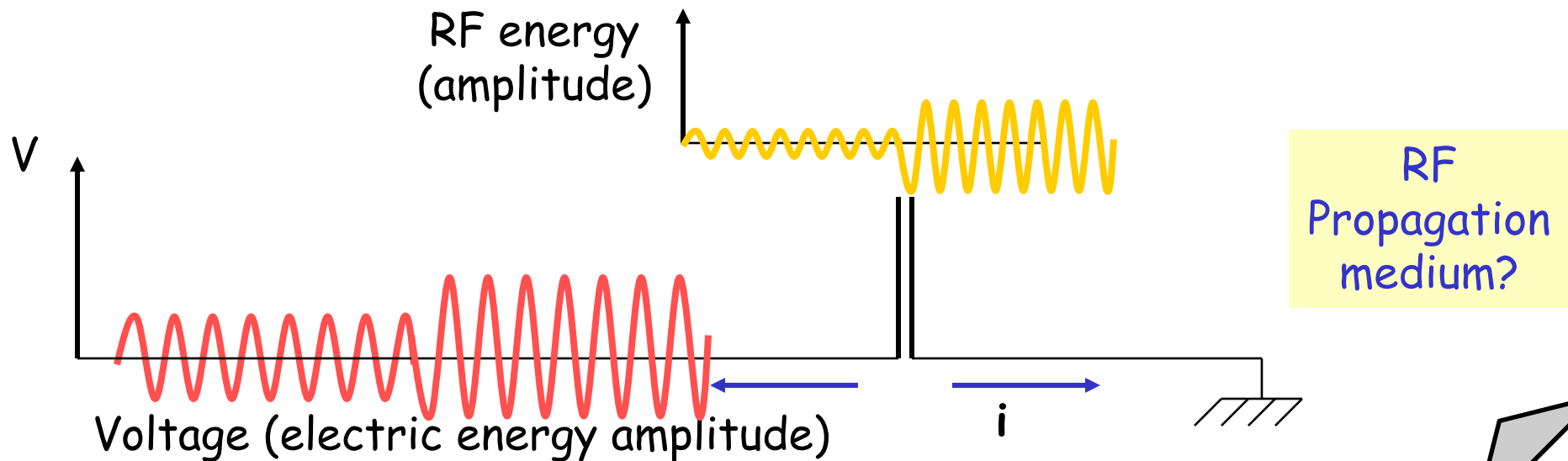
- Electromagnetic energy generated by high frequency alternate current (AC) in antennas
- Antenna: converts the wired current to RF and viceversa



RF Properties

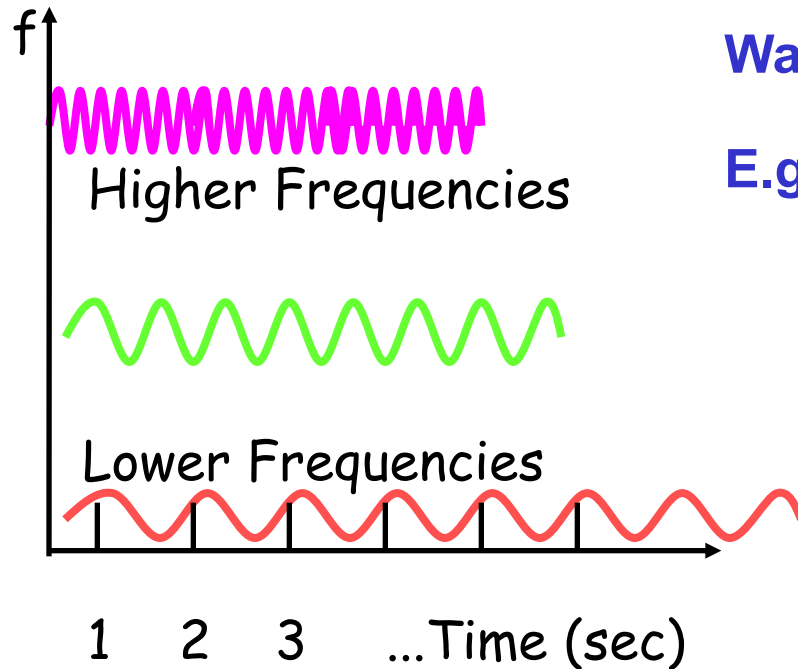
■ Amplitude

- Higher amplitude RF signals go farther
- Transmission Power (Watts) = Energy / Time = Joule / Sec
 - More energy (voltage) moves more electrons (current)
 - Power = Voltage * Current



RF Properties

- **Frequency (and Wavelength)**
 - Wireless Spectrum (see next slides)
 - Portion of wireless spectrum regulated by regional authorities and assigned to wireless technologies



Wavelength = c / frequency

E.g. 2.4 GHz (ISM band)



Wave Length =

$$300.000.000(\text{m/s}) / 2.400.000.000 \text{ Hz} = 0.125 \text{ m} = 12.5 \text{ cm}$$

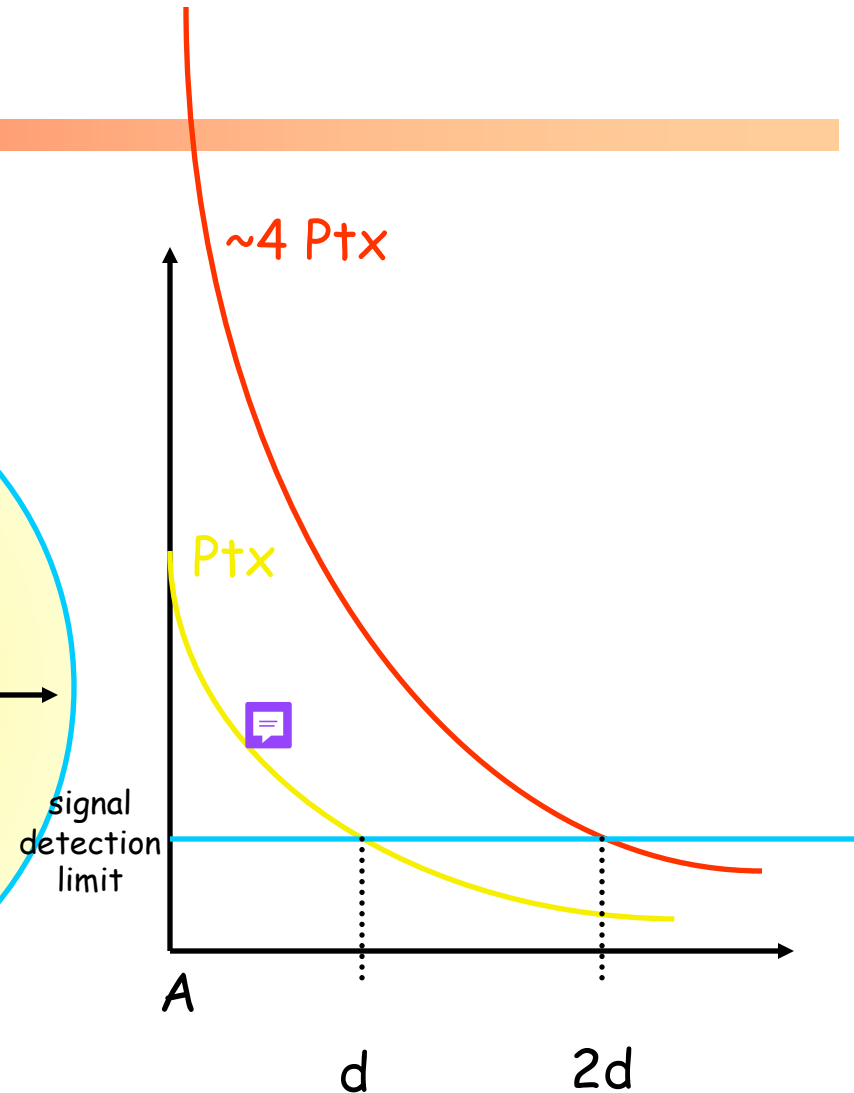
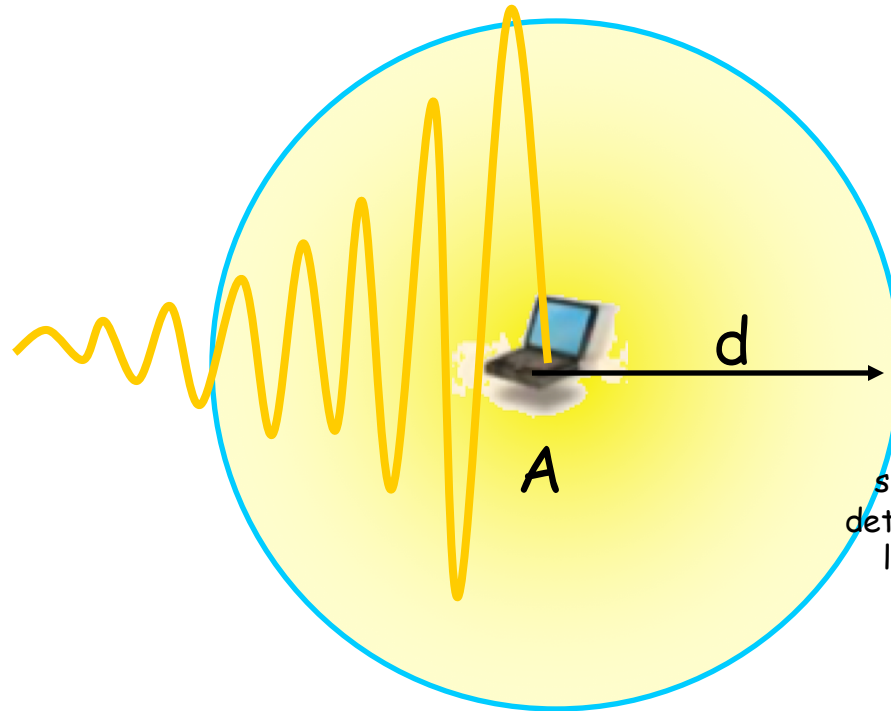
In practice:

Antennas work better
with size = $1, \frac{1}{2}, \frac{1}{4}$ of wavelegth
(try to measure antenna size of
your IEEE 802.11 device)



RF propagation

- Radio transmission coverage



The range is a function of power transmission (P_{tx})
Signal strength reduces with d^k
($k \geq 2..3$, no obstacles, isotropic radiator)

In 3D, sphere:
 $V = (4 \pi r^3 / 3)$
 $S = (4 \pi r^2)$

Wireless signal propagation ranges

■ Transmission range

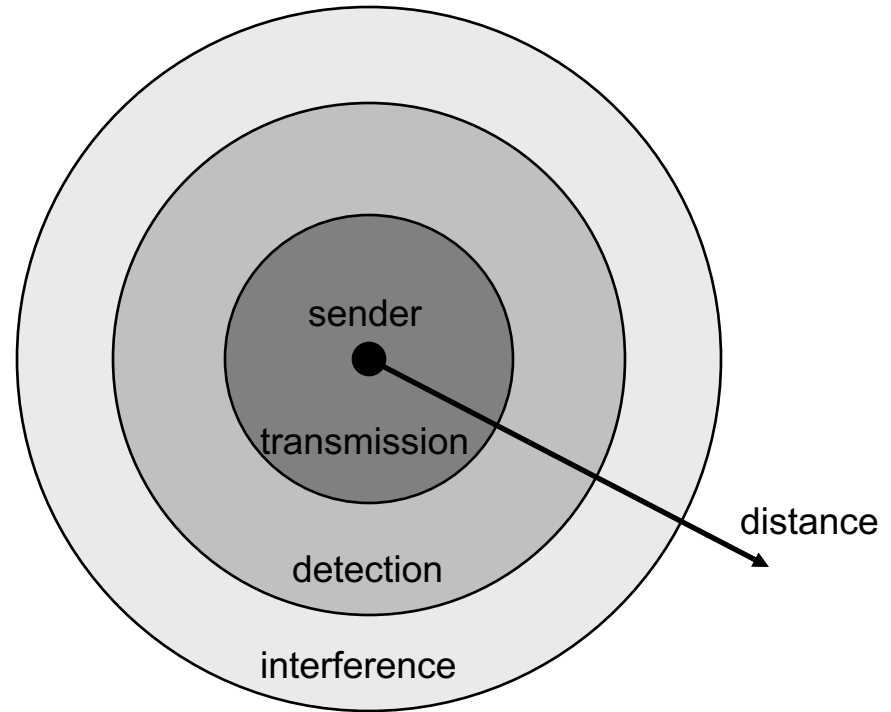
- communication possible
- low error rate

■ Detection range

- detection of the signal possible
- no communication possible

■ Interference range

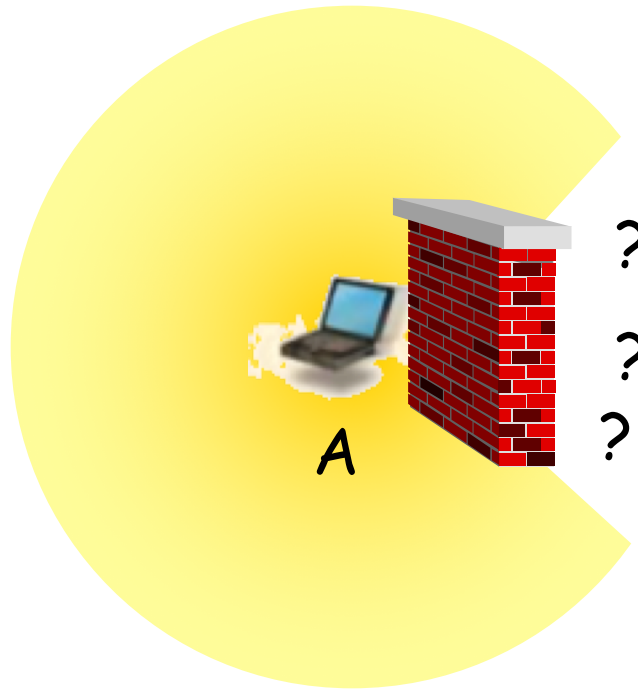
- signal may not be detected
- signal adds to the background noise



Ranges depend on receiver's sensitivity!

Wireless networks' technology

■ Radio transmission coverage



Rules of thumb:

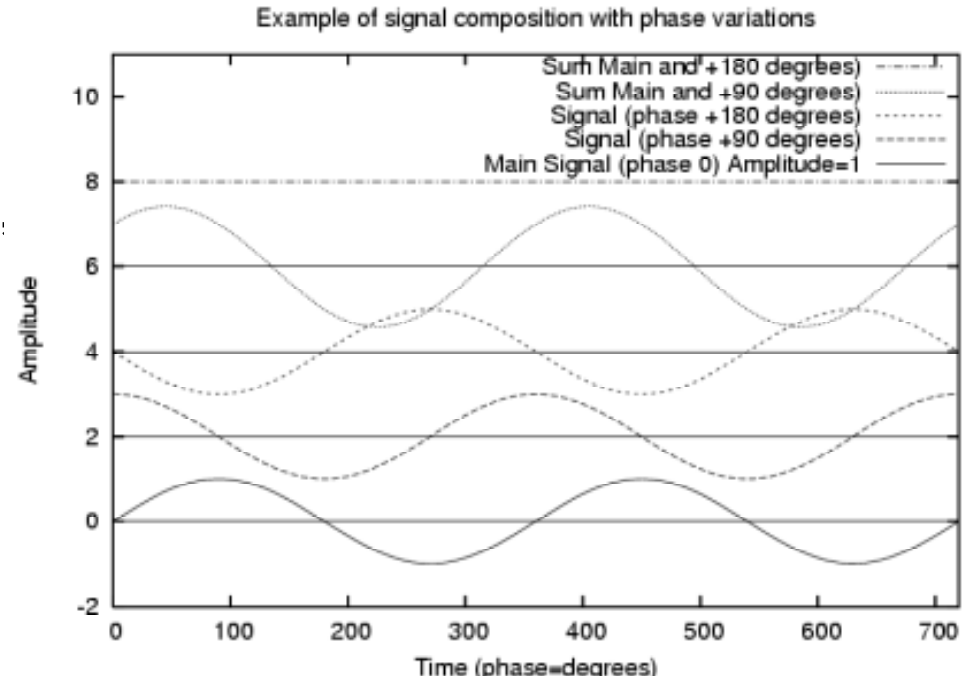
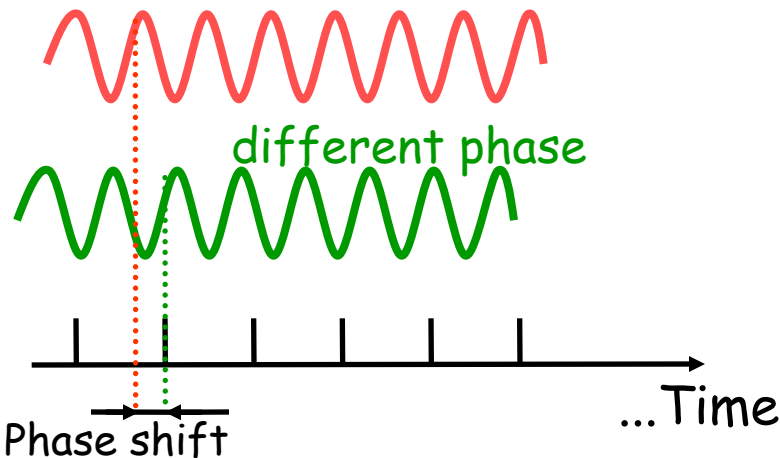
- high frequencies are good for short distances and are affected by obstacles
- low frequencies are good for long distances and are less affected by obstacles

obstacles can reflect or absorb waves depending on materials and wave frequencies

RF Properties

■ Phase: shift of the wave (in degrees or radians)

- Positive phase (left-shift), early wavefront
- Negative phase (right-shift), late wavefront



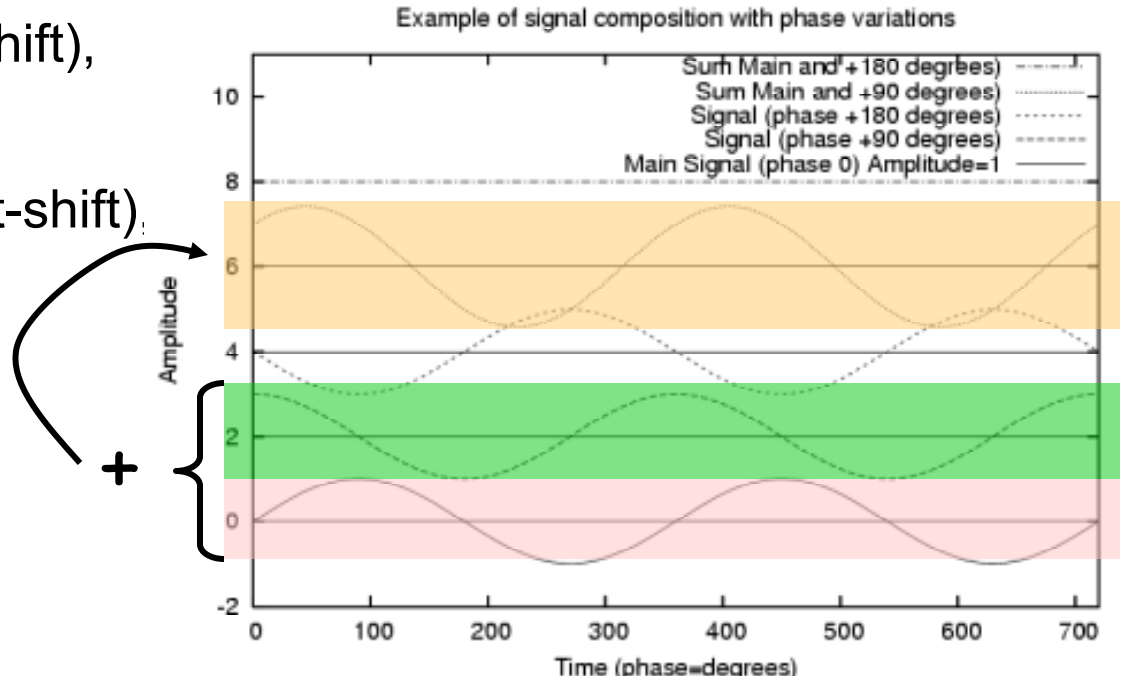
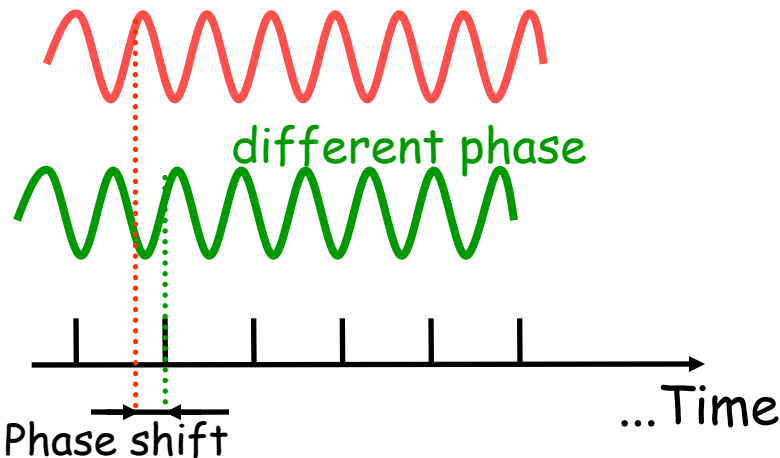
In practice:

RF echoes arriving at receivers with different phase may have positive or negative effects... Why?

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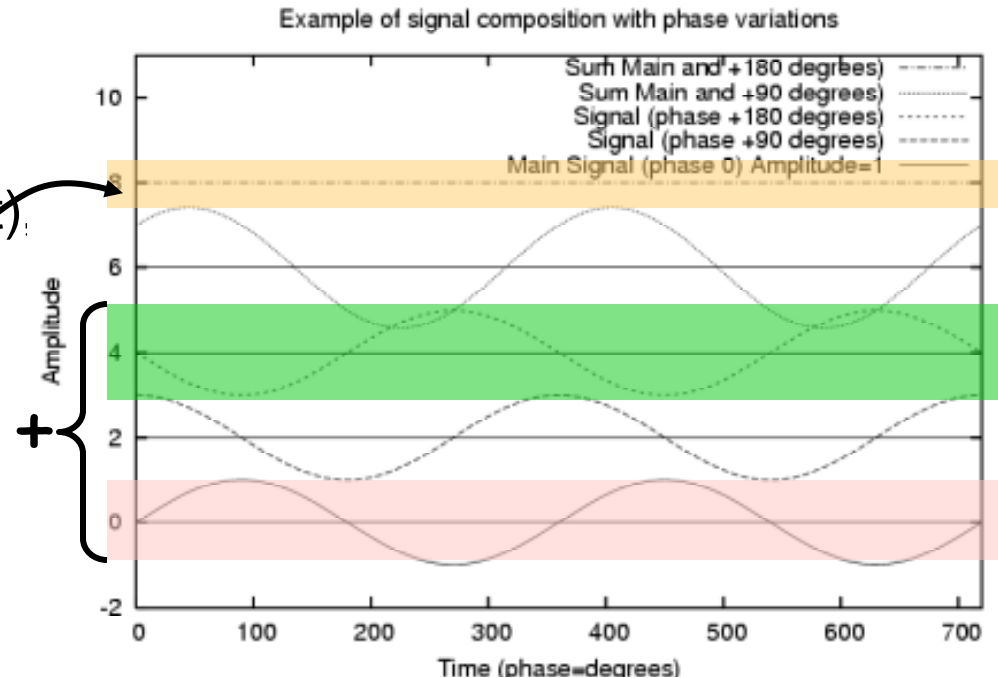
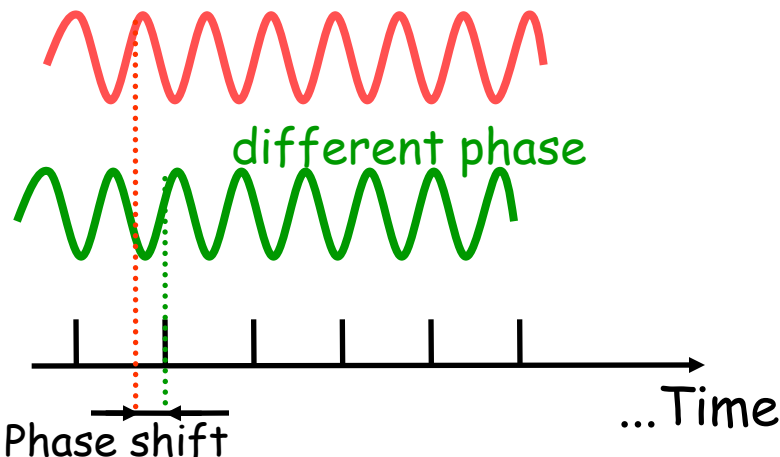
RF Properties

- **Phase: shift of the wave (in degrees or radians)**

- Positive phase (left-shift), early wavefront
- Negative phase (right-shift), late wavefront

Test with audio (demo):

<https://www.szynalski.com/tone-generator/>

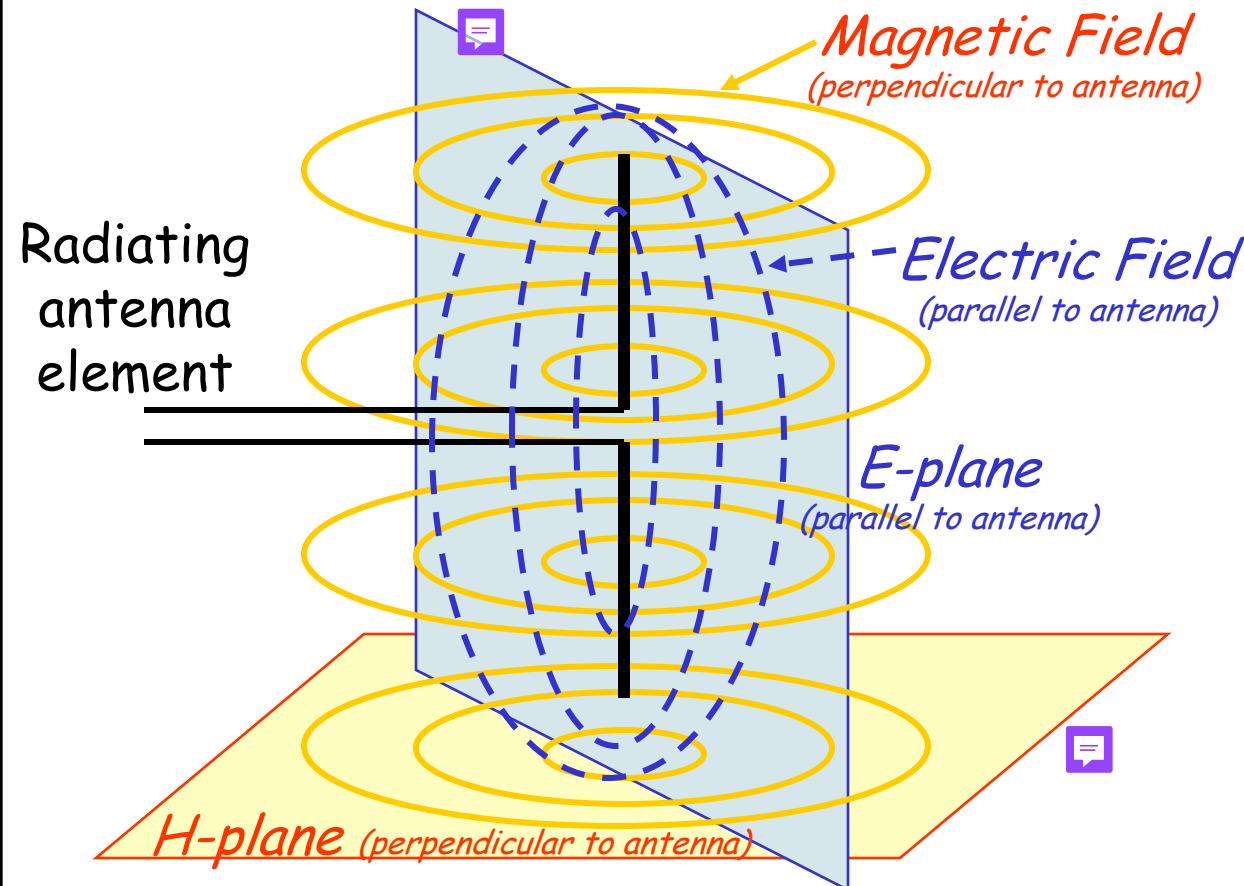


In practice:

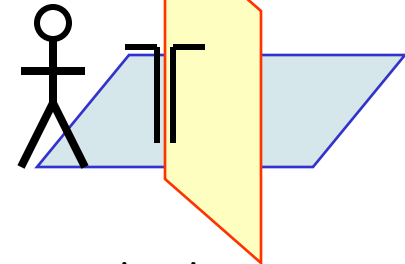
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RF Properties

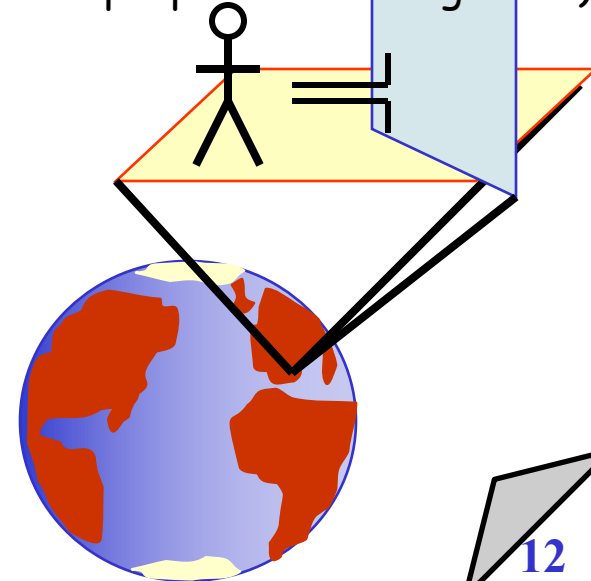
- **Polarization: (physical orientation of antenna)**
 - RF waves are made by two perpendicular fields:
 - Electric field and Magnetic field



Horizontal Polarization
(electric field is
parallel to ground)



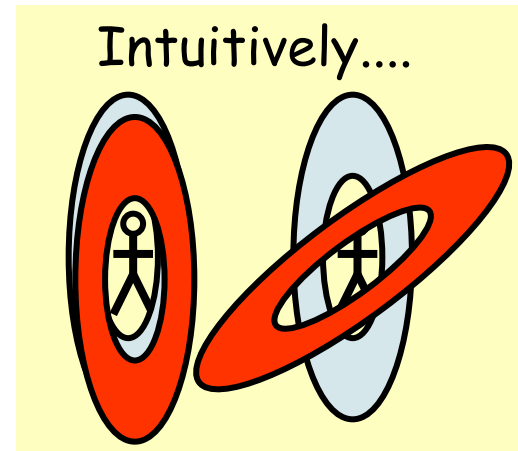
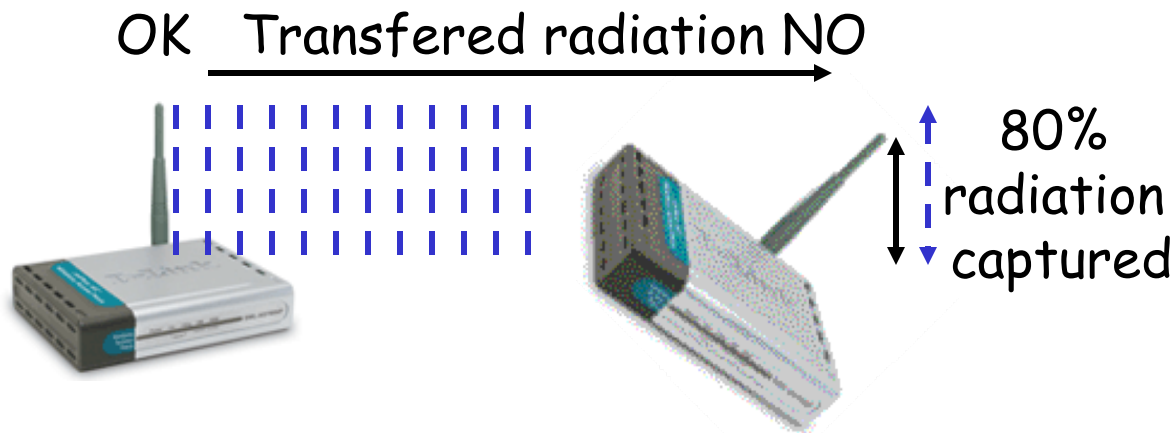
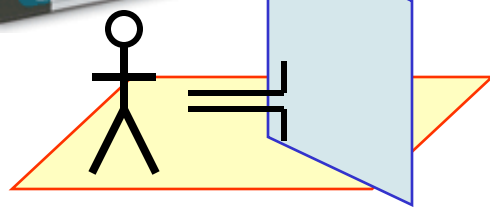
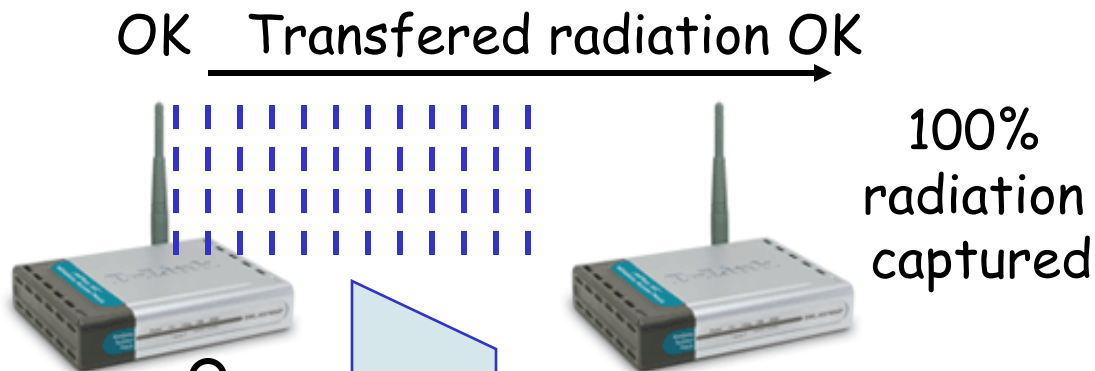
Vertical Polarization
(electric field is
perpendicular to ground)



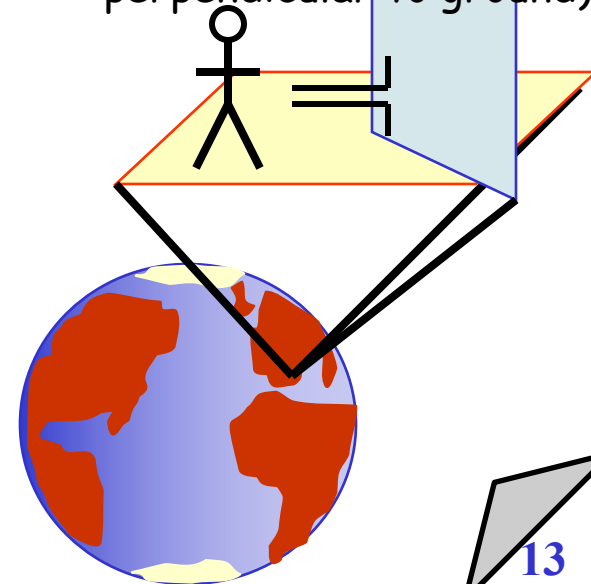
RF Properties



Vertical Polarization: typically used in WLANs

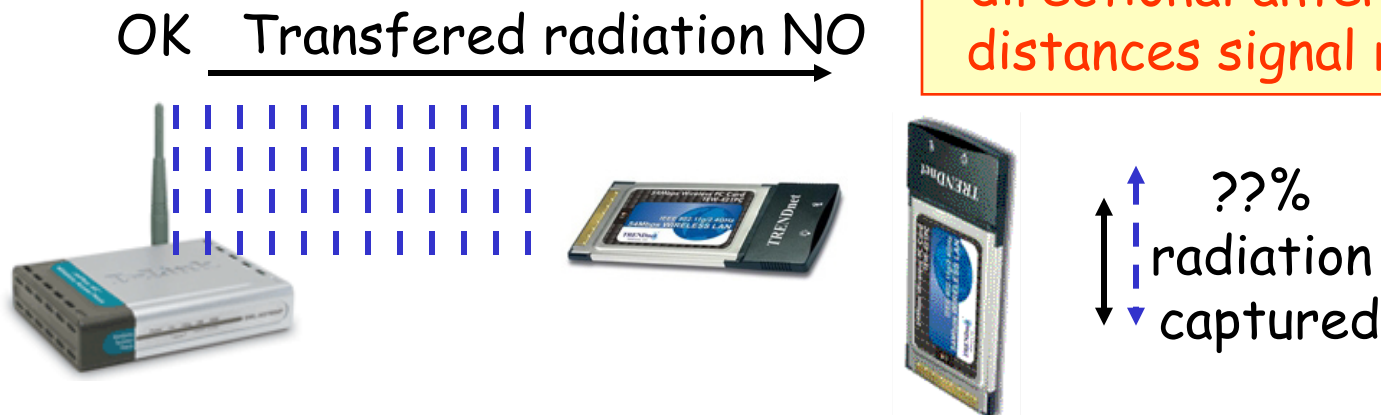
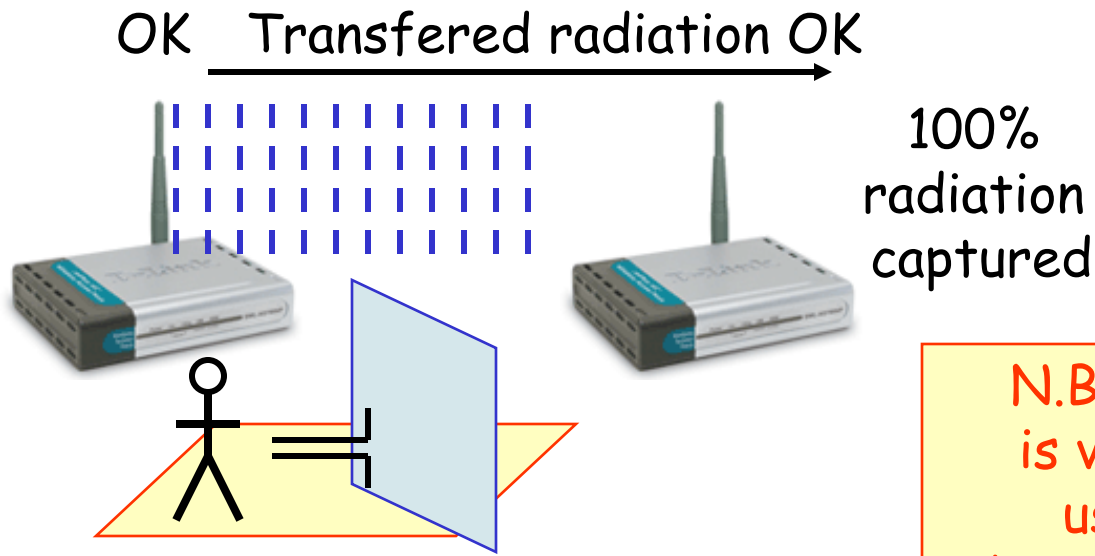


Vertical Polarization
(electric field is
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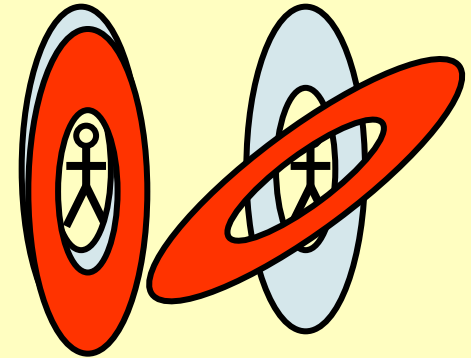


RF Properties

Vertical Polarization: the PCMCIA device problem



Intuitively....

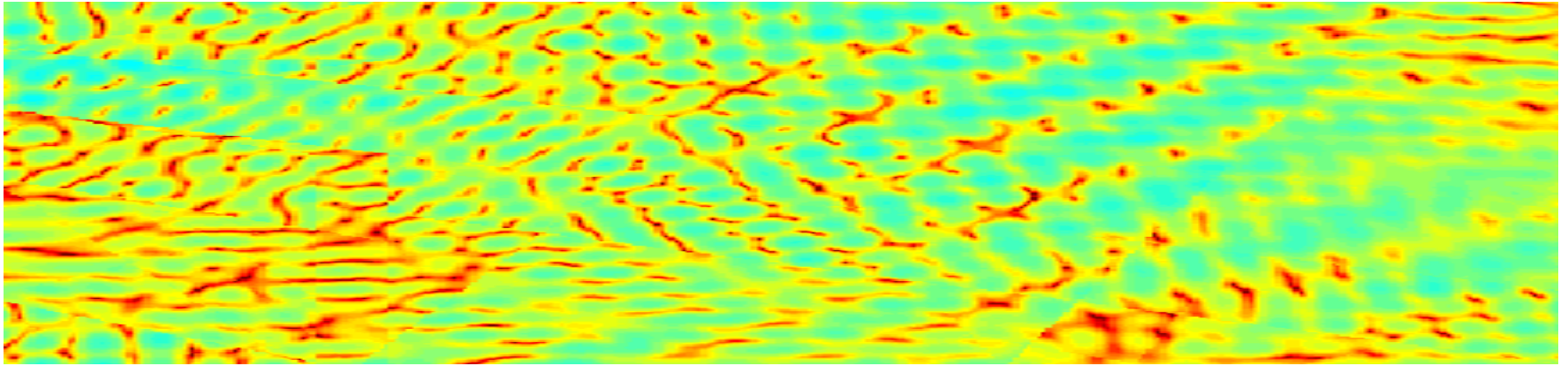
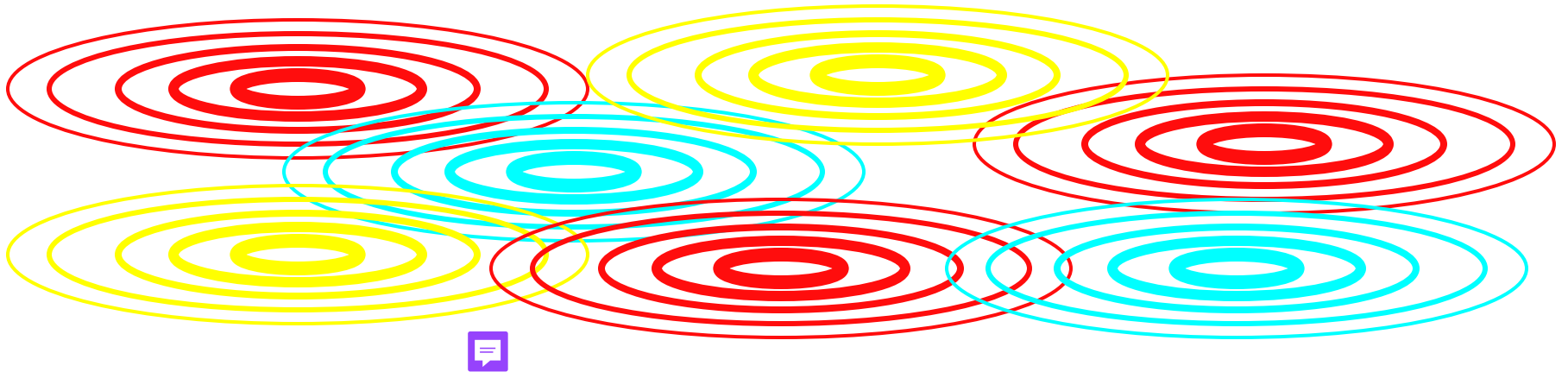


N.B. the polarization problem is very much important when using distant devices and directional antennas. With short distances signal reflections help!

RF Behaviors

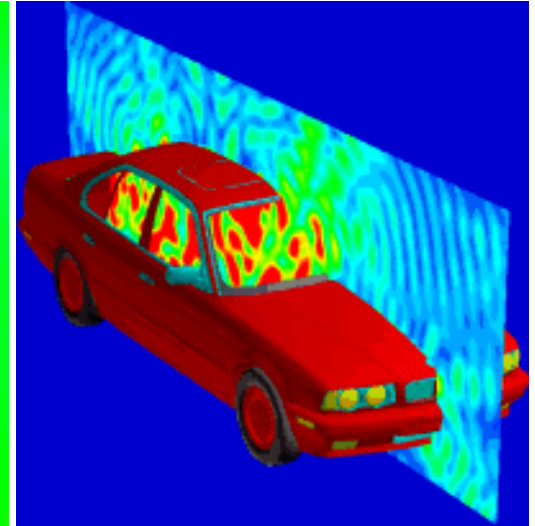
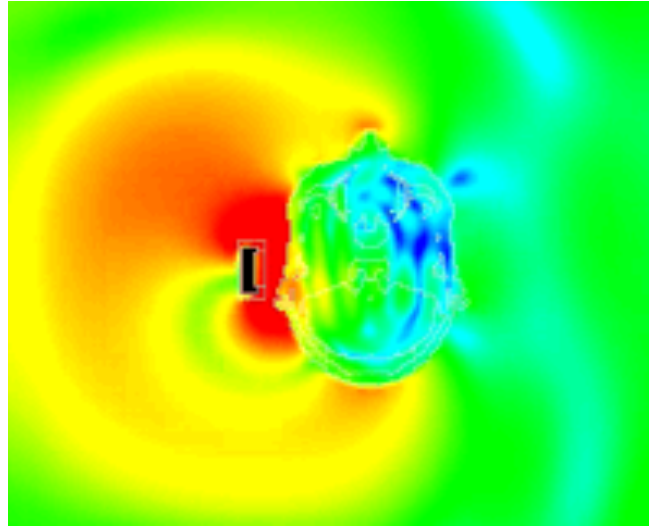
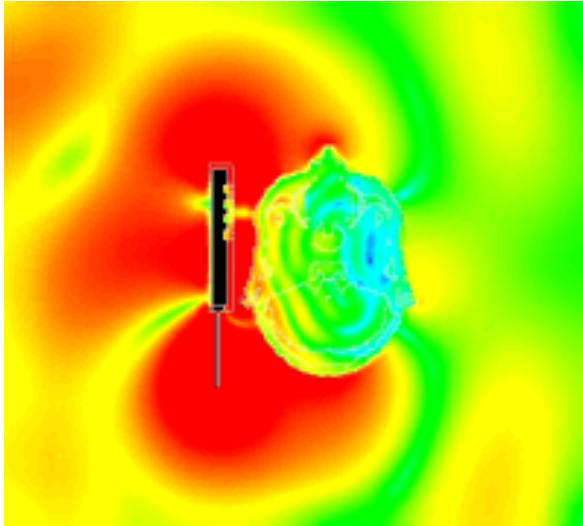
- Radio transmission interference

<http://www.met.rdg.ac.uk/clouds/maxwell/>



RF Behaviors

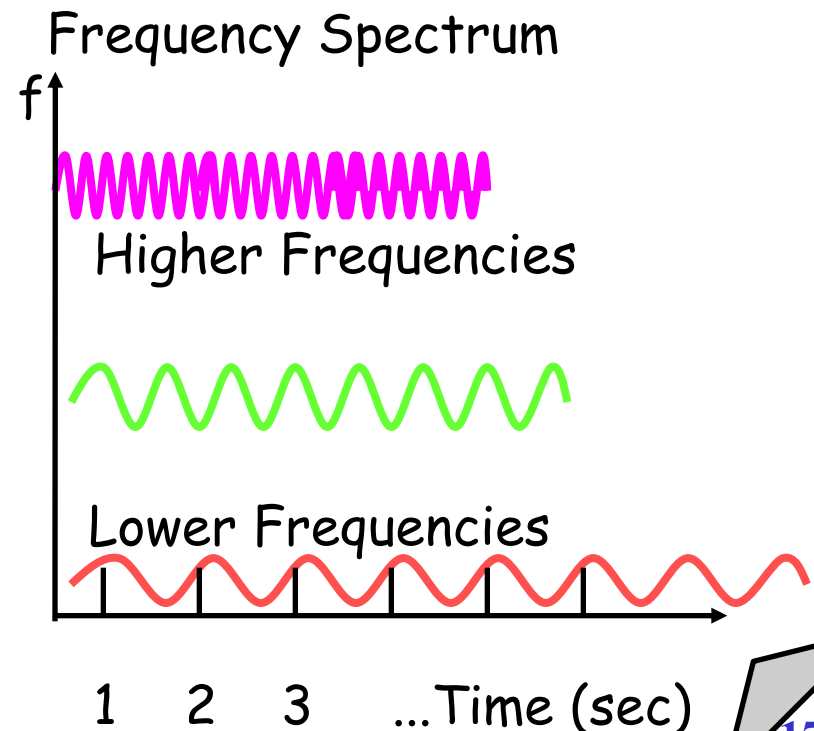
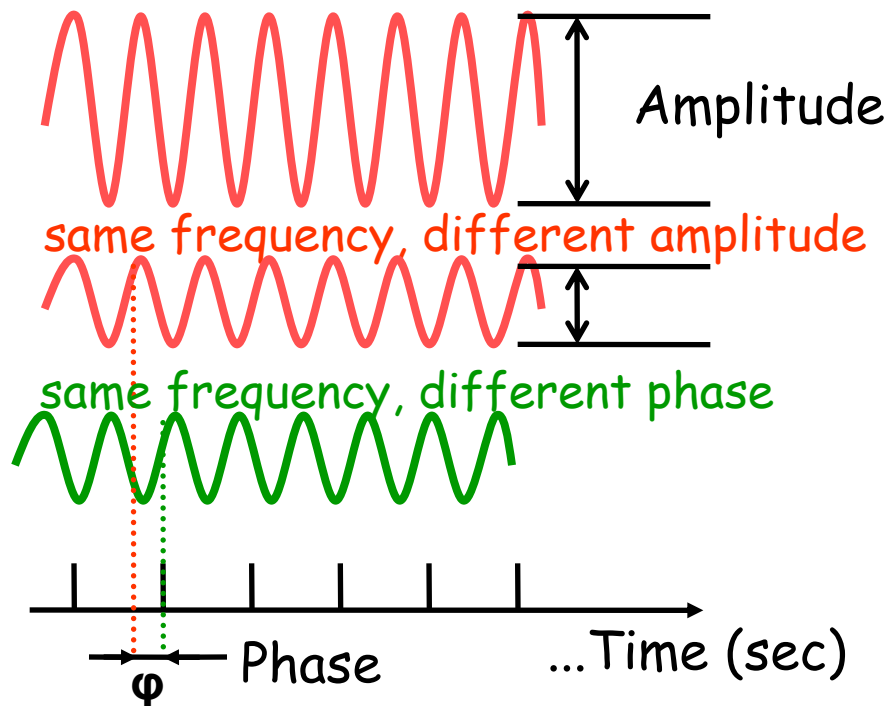
- Radio effects on human head (do not try this at home ☺)
<http://temf.de/Radiation-of-a-mobile-phone-P.58.0.html?&L=1>
- Credits: Technische Universitat Darmstadt, Computational Electromagnetics Laboratory



Wireless transmission: Electromagnetic waves

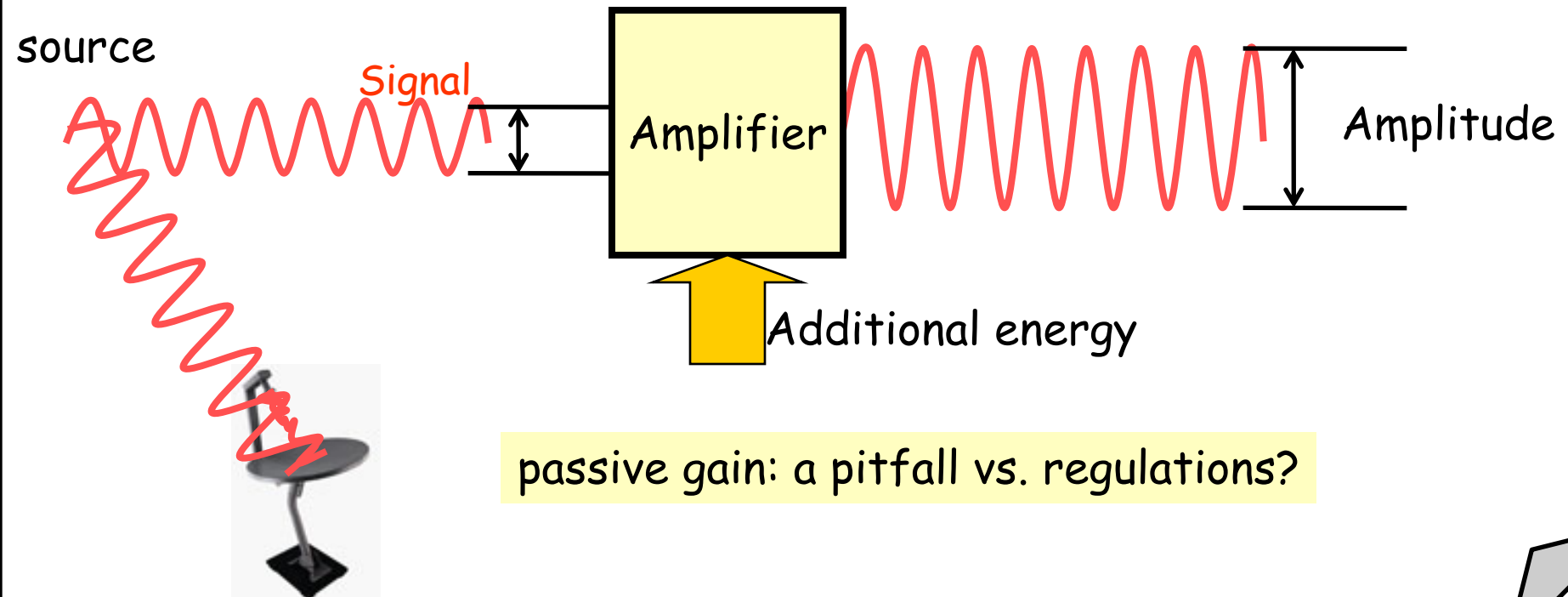
▪ Different parameters of electromagnetic waves:

- amplitude **M** proportional to transmission energy (loudness)
- frequency **f** (tone) measured in Hertz (Cycle/sec)
- phase **ϕ** (peak shift with respect to reference signal) (rad)



Wireless transmission

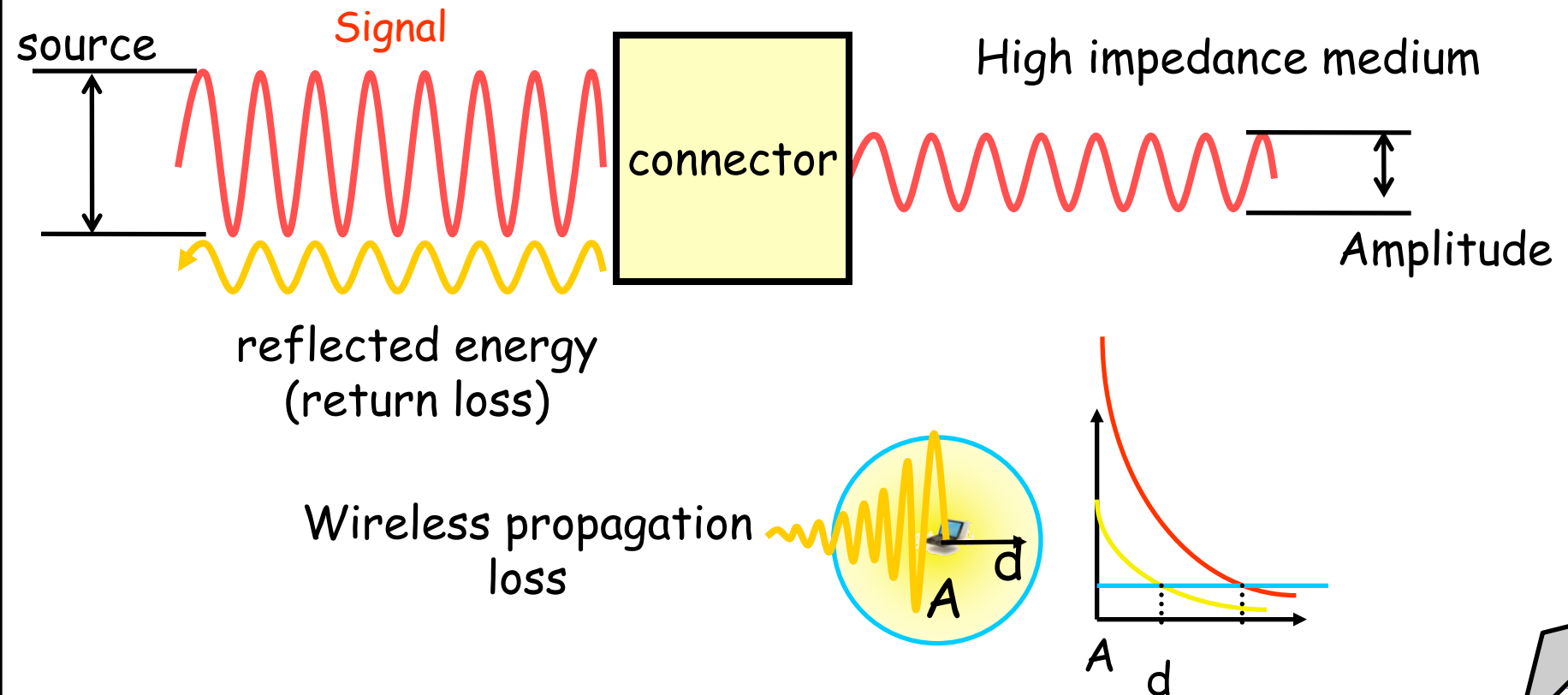
- **Signal Gain: (measured in Decibels, Db)**
 - Increase in amplitude **M** proportional to transmission energy
 - Active gain (amplifiers)
 - Passive gain (antennas focusing signal energy, and additive signal effects) 



Wireless transmission

▪ Signal Loss: (Db)

- Decrease in amplitude **M** proportional to energy waste
 - Intentional (resistance, signal attenuation -> heat)
 - Obstacles, e.g. (walls, water for 2.4 Ghz) and distance (wireless)



Wireless signal propagation ranges (reprise)

■ Transmission range

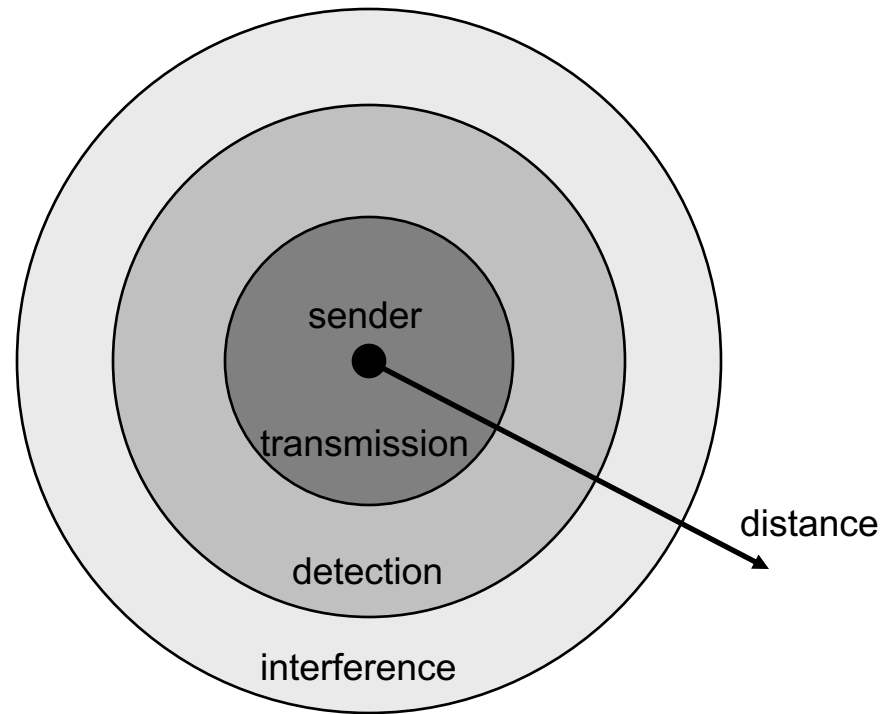
- communication possible
- low error rate

■ Detection range

- detection of the signal possible
- no communication possible

■ Interference range

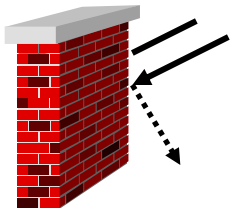
- signal may not be detected
- signal adds to the background noise



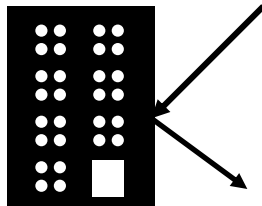
Ranges depend on receiver's sensitivity!

Wireless Signal propagation effects

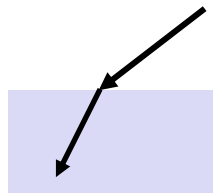
- Propagation in free space always like light (straight line)
- Receiving power proportional to $1/d^2$
(d = distance between sender and receiver)
- Receiving power additionally influenced by
 - fading (frequency dependent)
 - shadowing 🗨️
 - reflection at large obstacles
 - refraction depending on the density of a medium
 - scattering at small obstacles
 - diffraction at edges



shadowing



reflection



refraction

Reti di Calcolatori

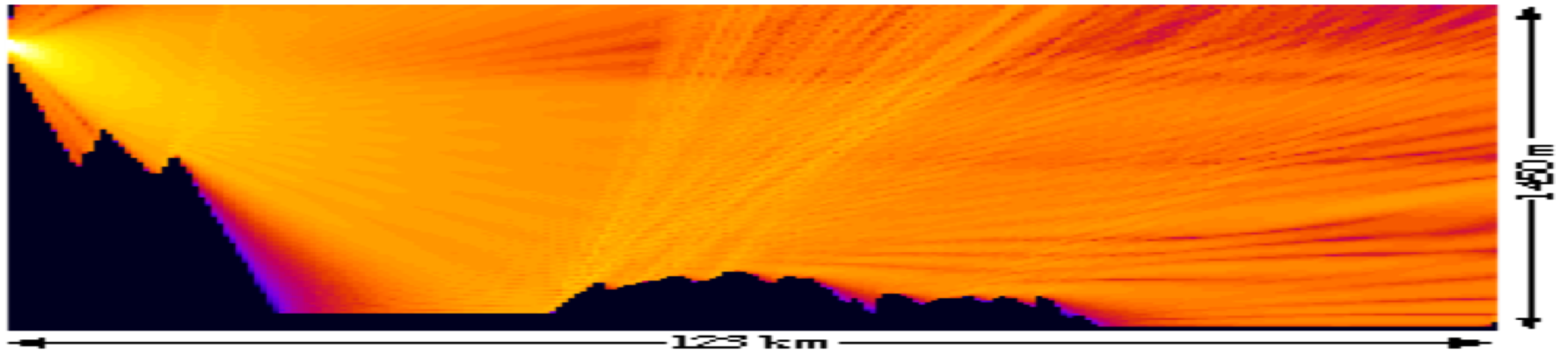


scattering



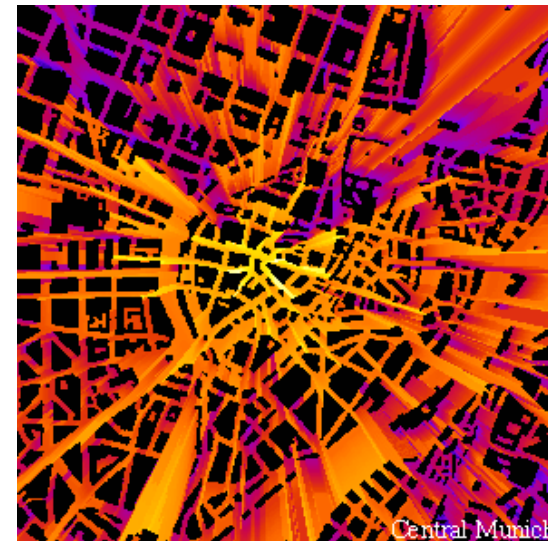
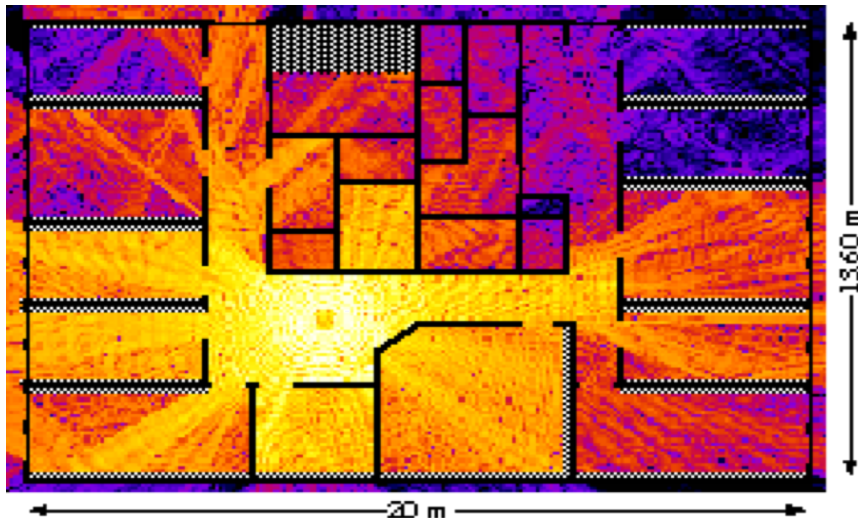
diffraction

Real world example



Raytracing examples

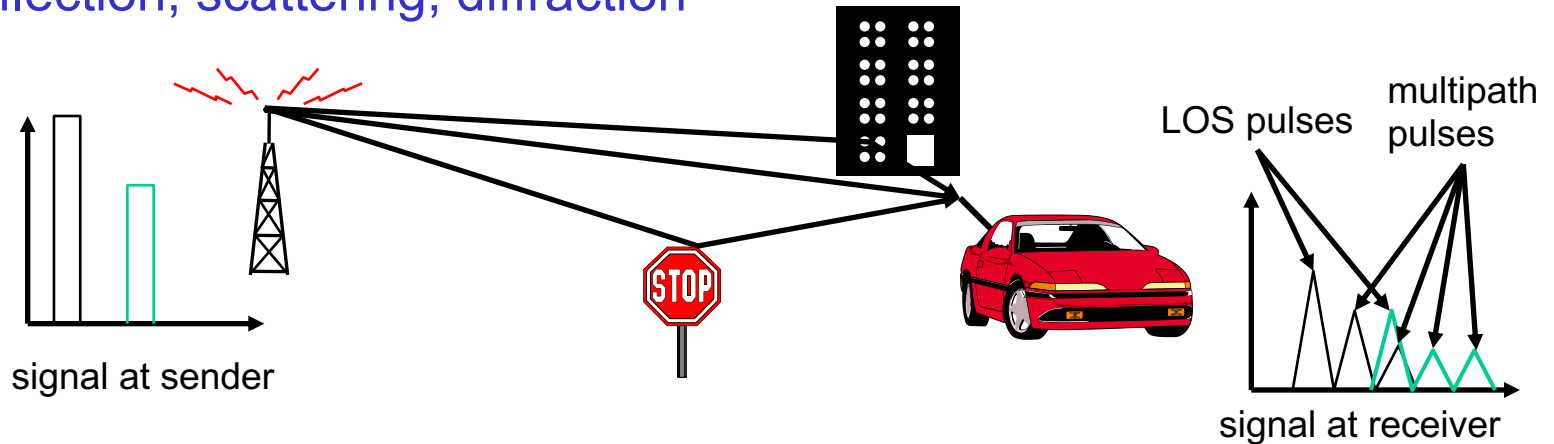
Low signal



high signal

Multipath propagation

- Signal can take many different paths between sender and receiver due to reflection, scattering, diffraction

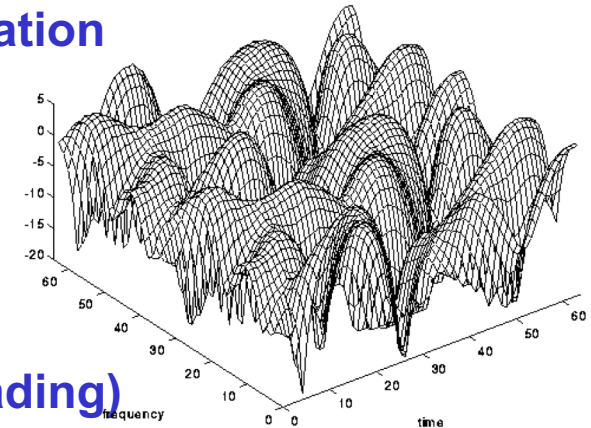


- Time dispersion: signal is dispersed over time
→ interference with “neighbor” symbols, Inter Symbol Interference (ISI)
- The signal reaches a receiver directly and phase shifted
→ distorted signal depending on the phases of the different parts

Effects of mobility

■ Channel characteristics change over time and location

- signal paths change
- different delay variations of different signal parts
- different phases of signal parts

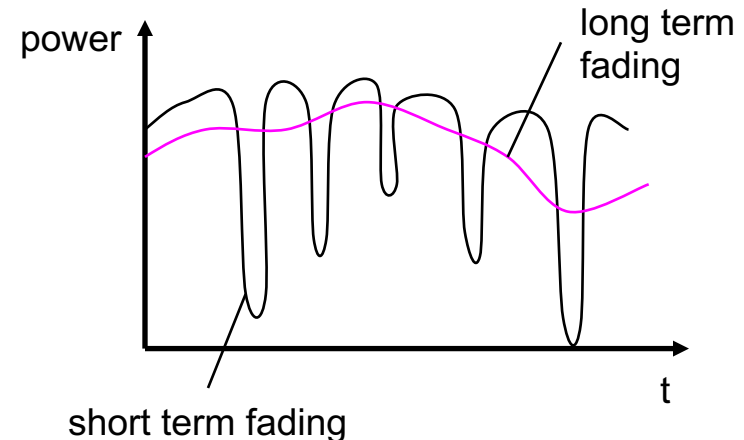


➔ quick changes in the power received (short term fading)

<http://www.sps.ele.tue.nl/members/j.p.linnartz/web/reference/chaptr03/rayjava/rayjava.htm>

■ Additional changes in

- distance to sender
- obstacles further away

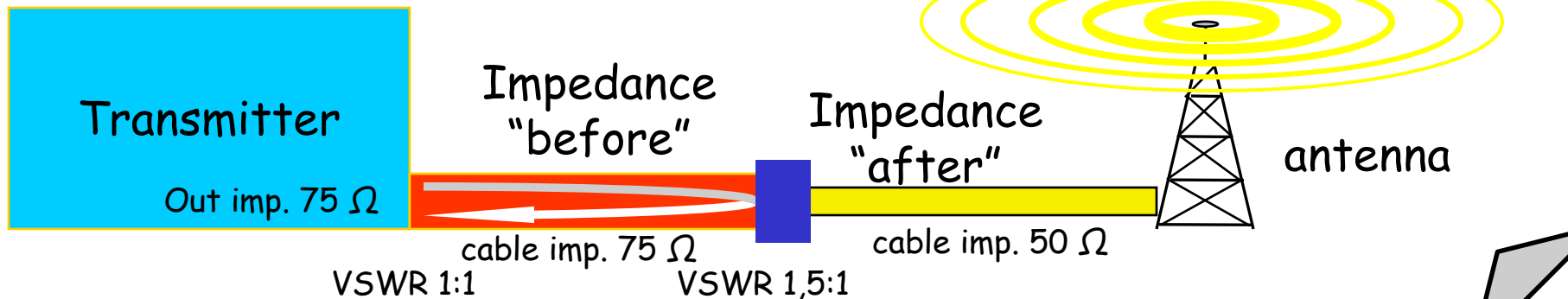


➔ slow changes in the average power received (long term fading)

Voltage Standing Wave Ratio (VSWR)

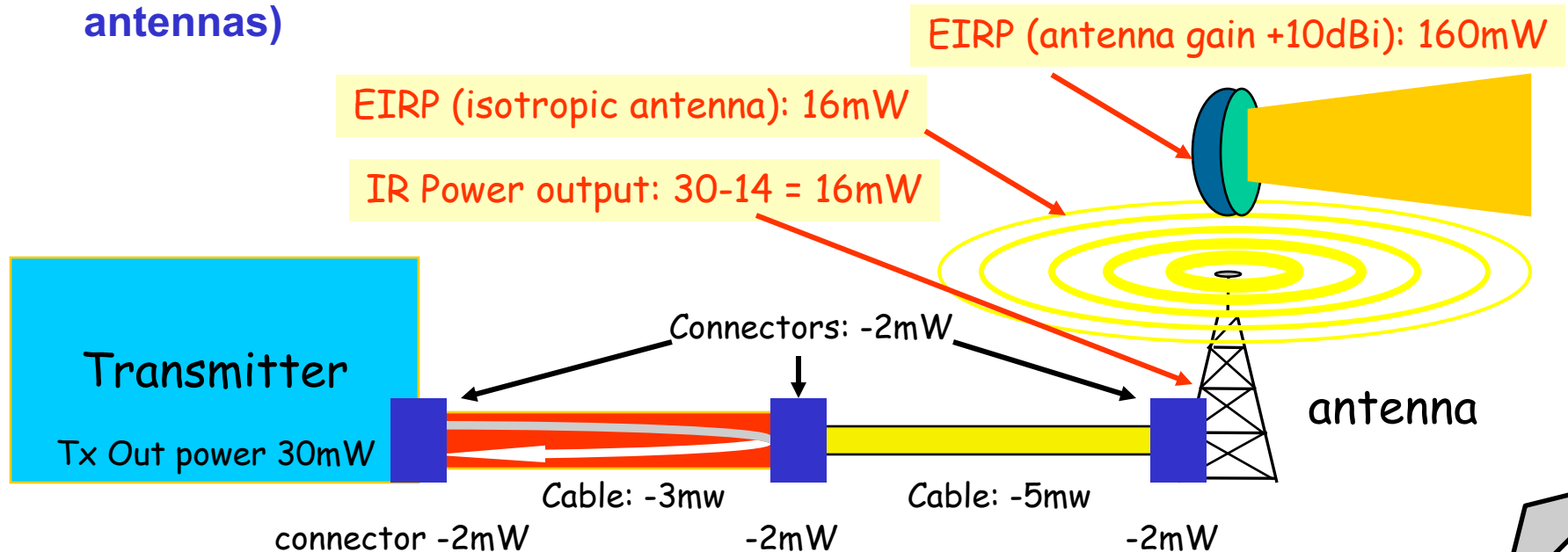
- **VSWR occurs with different impedance (Ohm) = resistance to AC current flow between transmitter and antenna**
 - VSWR is the cause of “return loss” energy towards the transmitter
 - Measured as ratio between impedance (before and after)
 - E.g. 1,5:1 (impedance ratio before/after is 1,5 times the ideal value)
 - 1 = normalized ideal impedance (1:1 means perfect VSWR)
 - VSWR Causes burnout of transmitter circuits, and unstable output levels

VSWR solution:
always use same impedance
circuits, cables, connectors
(typical 50 Ω in LANs)



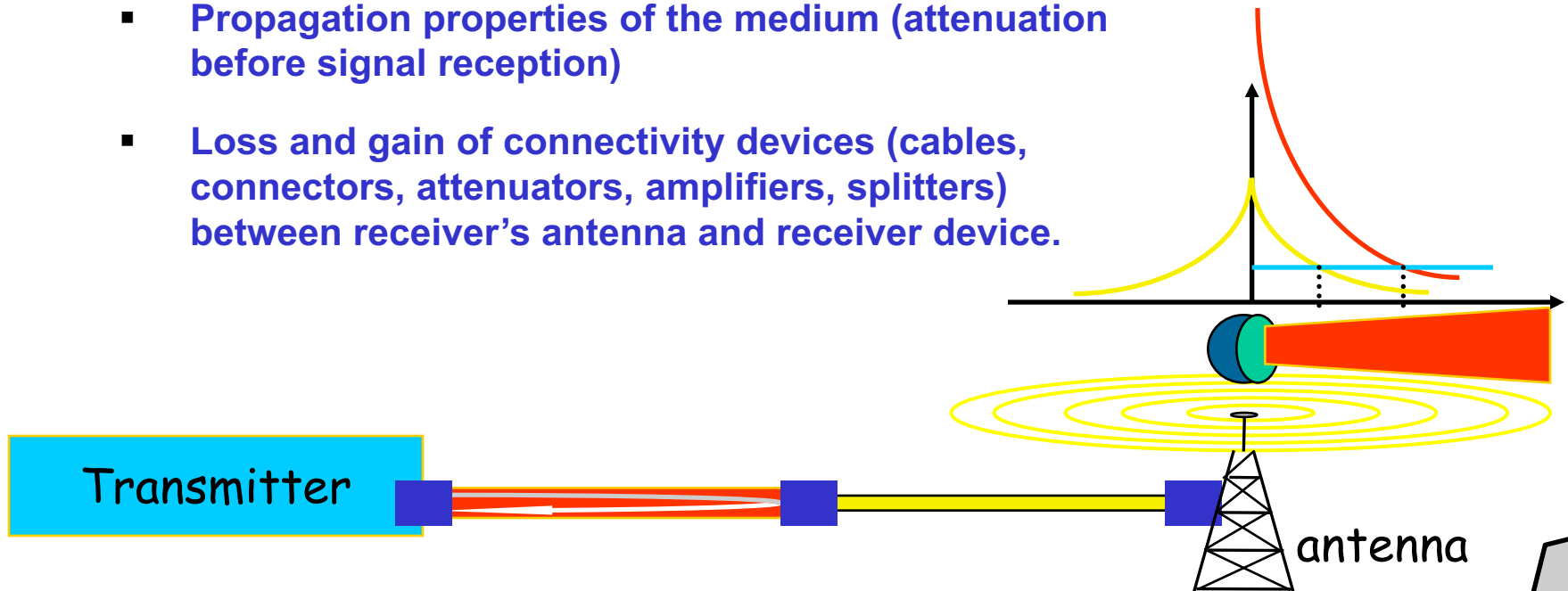
Intentional radiator and EIRP

- (Intentional) radiator: (def.) RF device specifically designed to generate and radiate RF signals.
 - ...Includes Tx RF device, cables and connectors (antenna excluded)
 - IR Power output: (subject to regulations) is the power output of last connector just before the antenna
- Equivalent Isotropically Radiated Power (EIRP): the power radiated by the antenna (including the passive antenna gain effect of directional antennas)



System design (under power viewpoint)

- Many factors must be considered in the design of Wireless systems:
 - Power of transmitting device
 - Loss and gain of connectivity devices (cables, connectors, attenuators, amplifiers, splitters) between transmission device and transmitter's antenna
 - Power of the intentional radiator (last connector just before antenna)
 - Power radiated by antenna element (EIRP)
 - Propagation properties of the medium (attenuation before signal reception)
 - Loss and gain of connectivity devices (cables, connectors, attenuators, amplifiers, splitters) between receiver's antenna and receiver device.



Power measurement

- **WATT: electric power unit**
 - **1 Watt = 1 Ampere * 1 Volt ($P=V*I$) also $P= R*I^2$ and $P = L/t$**
 - **Current (ampere) is the amount of charge (electrons) flowing as current in a wire**
 - **Voltage (Volt) is the “pressure” applied to the flow of charge**
 - **Resistance (impedance) is the obstacle to current flow**
 - **Power is the energy needed (in a given time unit) to apply a given “pressure” to a given “amount of charge”, by resulting in a flow of current.**
 - **Watt and dBm are units used for absolute power measurement**
 - **Typical RF power for WLANs:**
 - **AP: 30..100 mW (up to 250 mW outdoor), PCMCIA: 15..30 mW**

Power measurement

- **Decibel (dB):** a power measurement unit designed to express power loss
 - It is more practical to use given the logarithmic decay of wireless signals
 - It allows to make easy calculations on “resulting power”
- **Decibel (dB) measures the logarithmic relative strength between two signals (mW are a linear absolute measure a energy)**

- $\text{Log}_{10}(X) = Y \quad \Leftrightarrow \quad 10^Y = X$

Exponential
growth

- $1 = 10^0, \log_{10}(1) = 0$

- $10 = 10^1, \log_{10}(10) = 1$

- $100 = 10^2, \log_{10}(100) = 2$

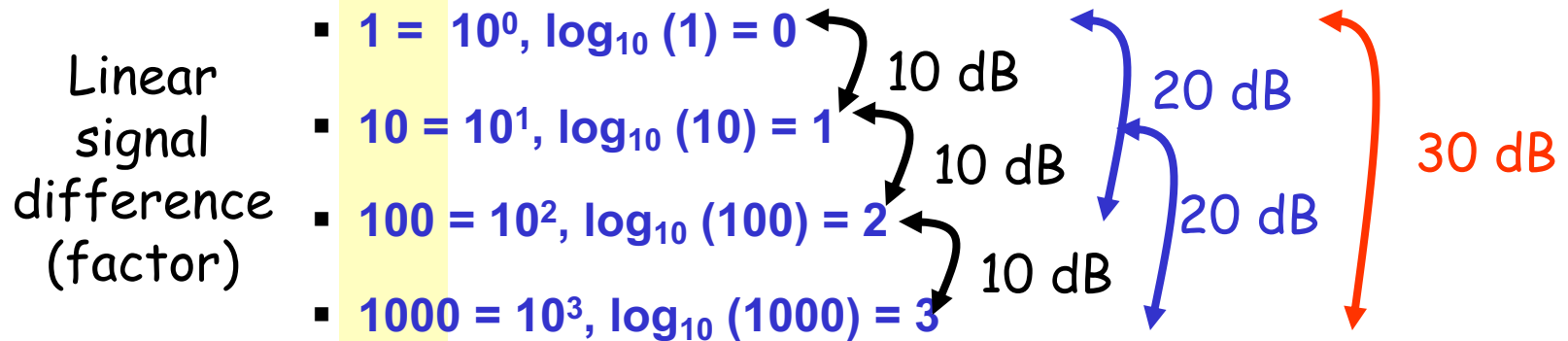
- $1000 = 10^3, \log_{10}(1000) = 3$

Linear growth
“BEL” units (B)

- **How strong is a 10 dB signal? (it depends on the reference signal)**

Power measurement

- Decibel (dB): 1/10 of a Bel
- E.g. 1000 is one Bel greater than 100 $\Rightarrow$ 1000 is 10 dB greater than 100



- How strong is a 10 dB signal? (it depends on the reference signal)
 - Positive dB value is power gain, negative dB value is power loss
 - e.g. given 7 mW power, a +10 dB signal gain is 70 mW
 - e.g. given 7 mW power, a -10 dB signal gain (loss) is 0.7 mW
- Power Difference (in dB) between Tx and Rx signal:
 - Power Difference (dB) = $10 * \log(\text{Power Rx(Watt)} / \text{Power Tx (Watt)})$
- Gain and Loss are relative power measurements: dB is the unit

Power measurement

- Advantage of dB: what is better?

- E.g.: A signal transmitted at [TX] 100 mW is received at [RX] 0.000005 mW
 - Power Difference (dB) = $10 * \log([RX] / [TX]) = 10 * \log(0.000005\text{mW}/100\text{mW}) = -73$
 - A signal transmitted at 100 mW is received with gain (loss) -73 dB

- Advantage of dB: what is better?

- E.g.: A signal transmitted at 100 mW is received at 0.000005 mW, then it is amplified (*100) to 0.0005 mW ???
- A signal transmitted at 100 mW is received with gain (loss) $-73+20= -53$ dB

-3 dB	½ power in mW (/ 2)
+3 dB	2x power in mW (* 2)
-10 dB	1/10 power in mW (/ 10)
+10 dB	10x power in mW (* 10)

Approximated table (values defined for ease of calculations)

Power measurement

- **Practical example:**
 - Signal Tx at 100 mW, cable **-3dB** loss, amplifier **+10 dB** gain
 - $100 \text{ mW} / 2 \text{ (-3dB)} = 50 \text{ mW} * 10 \text{ (+10 dB)} = \mathbf{500 \text{ mW}}$ IR power output
 - Signal TX at **30 mW** is received at the antenna as **6 mW** (2/10 of TX power)
 - Intentional Radiator Gain (loss) = $30 \text{ mW} / \mathbf{10} = 3 \text{ mW} * \mathbf{2} = 6 \text{ mW}$
 - Intentional Radiator Gain (loss) = $-10 \text{ dB} + 3 \text{ dB} = \mathbf{-7 \text{ dB}}$ ($\approx 1/5$, $7 \text{ dB} \approx 5 \times$)
- **N.B. dBs are additive measures** of gain (loss): e.g. $6 \text{ dB} = +3+3 \text{ dB}$, $7 \text{ dB} = 10-3 \text{ dB}$
 - E.g. $100 \text{ mW} - 6 \text{ dB} = 100 \text{ mW} - 3 - 3 \text{ dB} = 100 / 2 / 2 = 25 \text{ mW}$
 - E.g. $100 \text{ mW} + 7 \text{ dB} = 100 \text{ mW} + 10 - 3 \text{ dB} = 100 * 10 / 2 = 500 \text{ mW}$
 - E.g. $10 \text{ mW} + 5 \text{ dB} = 10 \text{ mW} (+10+10-3-3-3-3-3) \text{ dB} = 1000/32 = 31.25 \text{ mW}$
 - E.g. $10 \text{ mW} + 11 \text{ dB} = ?$
 - E.g. $50 \text{ mW} - 8 \text{ dB} = ?$

N.B. Approximated values (values defined for ease of calculations)

Power measurement

- **dBm: dB-milliWatt, the absolute measure of signal power**
 - Assumption: reference signal is 1 mW = 0 dBm(normalization factor)
 - Useful for gain/loss calculation without passing through mW
 - E.g. access point transmits 100 mW = 1mW (*10*10) =+20 dBm
 - PCMCIA card transmits at 30 mW = 1mW (*10*3) = +14.7 dBm
 - E.g. Tx= 30 mW, cable –2 dB, amplifier +9 dB:
 - 30 mW = 1mW *10 *3 = 14.7 dBm
 - IR power : 14.7 dBm –2dB +9dB = 21.7 dBm (147.91 mW)
 - In general, for converting mW to dBm and viceversa:
 - $P_{\text{dBm}} = 10 \log(P_{\text{mW}})$ and $P_{\text{mW}} = 10^{(P_{\text{dBm}} / 10)}$

Power measurement

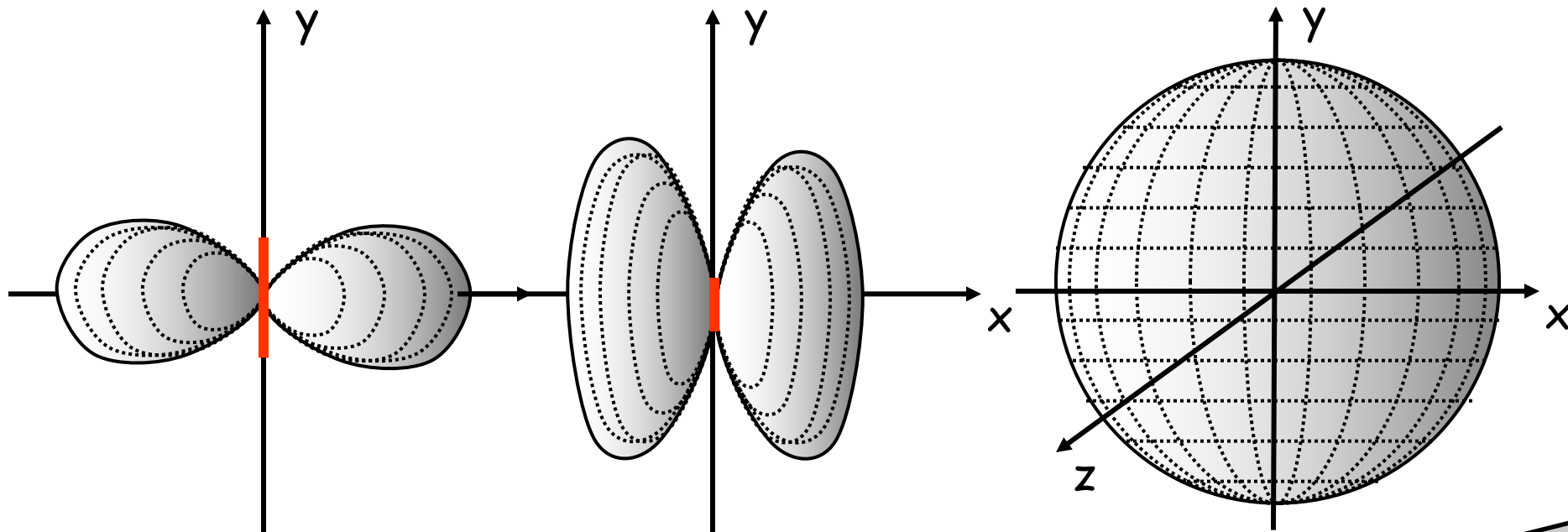
- mW - dBm: conversion table**

-40 dBm	-30 dBm	-20 dBm	-10 dBm	0 dBm	+10 dBm	+20 dBm	+30 dBm	+40 dBm
100 nW	1 μW	10 μW	100 μW	1 mW	10 mW	100 mW	1 W	10 W

-12 dBm	-9 dBm	-7 dBm	-6 dBm	-3 dBm	0 dBm	+3 dBm	+6 dBm	+7 dBm	+9 dBm	+12 dBm
62,5 μW	125 μW	200 μW	250 μW	500 μW	1 mW	2 mW	4 mW	5 mW	8 mW	16 mW

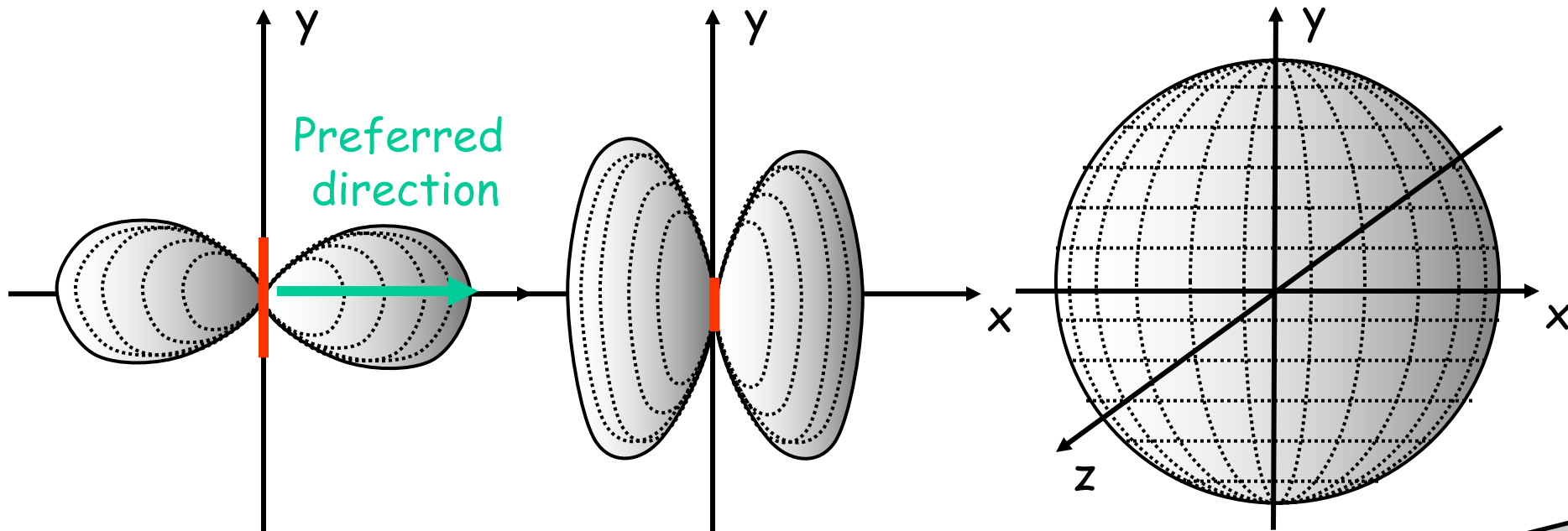
Power measurement

- **dBi: dB-isotropic**, the normalized measure of antenna passive gain
 - Assumption: an isotropic radiator has 100% efficiency in radiating energy in uniform way in every direction (e.g. the Sun)
 - Antennas concentrate energy in non-isotropic way, resulting in **passive gain (space dependent)**. Ideal antenna: zero length dipole



Power measurement

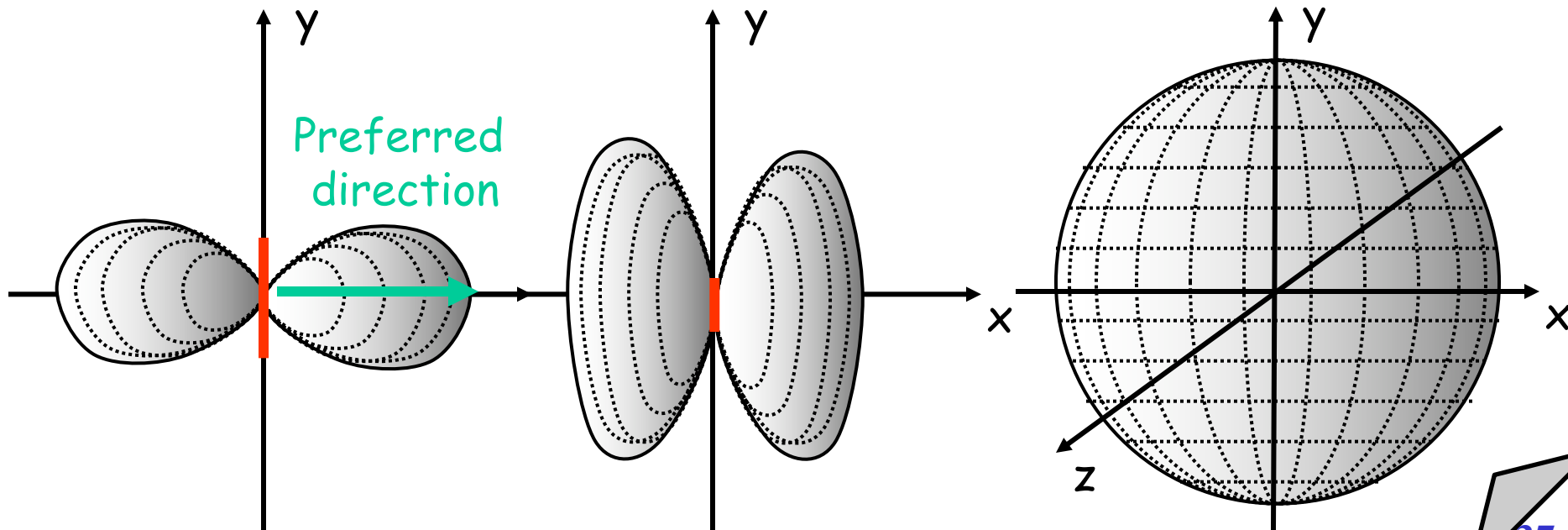
- **dB*i***: dB-isotropic, the normalized measure of antenna passive gain
 - If an antenna located in the origin $(0,0,0)$ has twice the radiated energy of an isotropic radiator in a given point (x,y,z) , then the antenna gain in (x,y,z) can be defined as +3 dBi. If the energy is 10x the isotropic radiator, the gain is +10 dBi, etc.etc.
 - Q: If the antenna gain is 7 dBi in (x,y,z) ?



Power measurement

- **dBi: dB-isotropic**, the normalized measure of antenna passive gain
 - Real antennas always have a preferred direction where the power is greater than isotropic radiator: **gain is always positive in the preferred direction!**
 - **Example: 1 mW IR power applied to directional antenna with +10 dBi gain in the preferred direction, would translate in EIRP?**
 - **EIRP = 1mW + 10 dBi = (10x) = 10 mW EIRP**

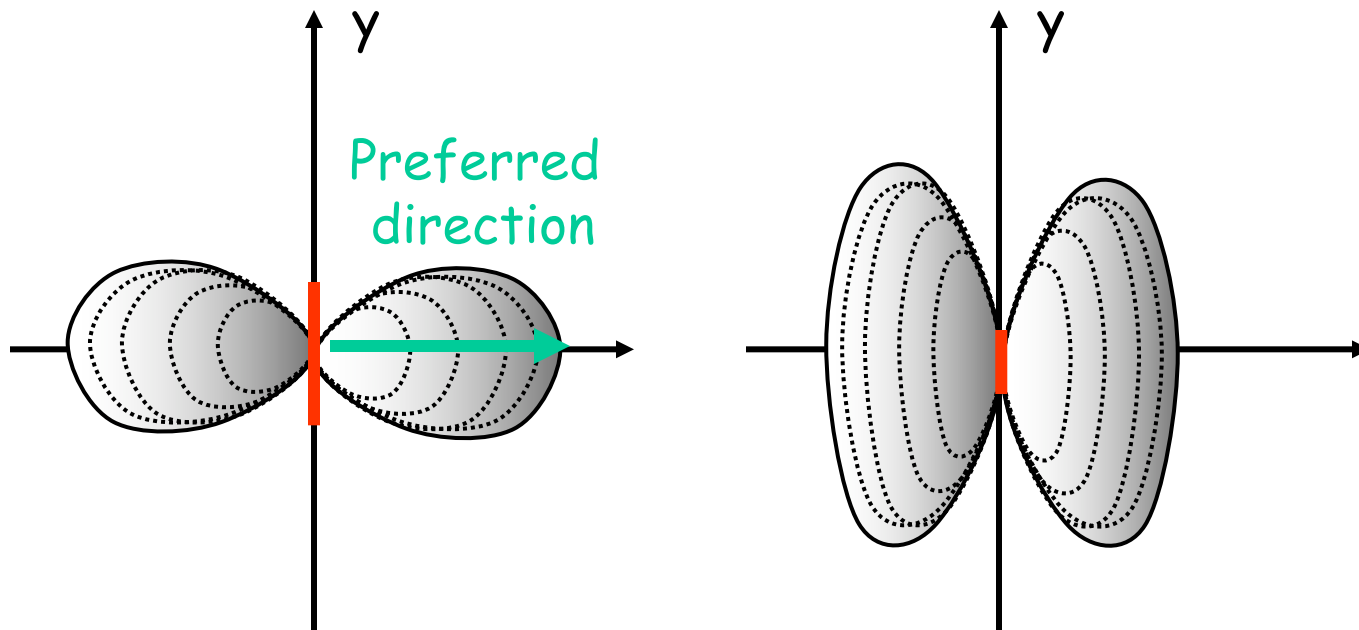
N.B. this does not mean that antenna generates more power !
Antenna concentrates power in preferred direction.



Power measurement

- **dBd: dB-dipole, the normalized antenna passive gain vs. 2,14 dBi half-wave dipole**
 - Reference is a half wave dipole with 2.14 dBi gain in **preferred direction**!
 - **Conversion rule:**
 - **0 dBd = 2.14 dBi, $\text{dBd} = (\text{dBi} - 2.14),$ $\text{dBi} = (\text{dBd} + 2.14)$**

Reference dipole



Power monitoring (e.g. IEEE 802.11 devices)

- (received) Power monitoring in IEEE 802.11 devices is needed for making device driver to work properly (typical sensitivity range is $[-90..+10]$ dBm):
 - Detect signal (below or above the sensitivity threshold?)
 - Detect signal power (selection of coding technique... That is bitrate!)
 - Detect channel status: idle? Ok, transmit! Busy? Ok, wait.
- Received Signal Strength Indicator (RSSI)
 - Index defined for IEEE 802.11 devices (check device analyzer, if any)
 - RSSI = function (dBm or mW received) = pure number reported to device driver!
 - Unfortunately the RSSI scale is not standard, that is, device dependent!
 - This fact does not allow to compare if device A receives better than device B (assuming different manufacturer) based on RSSI measurement
 - Problem: device A indicates maximum RSSI=255 (8 bits) with -10 dBm signal (0.1 mW), and device B indicates maximum RSSI=32 (5 bits) with -15 dBm (0.03 mW). Q: when both A and B in (x,y,z) receive -15 dBm, which one is better device? That is, which one would you buy if you are a system admin?



Antennas

- **Illustration of general issues**
 - **Convert electrical energy in RF waves (transmission), and RF waves in electrical energy (reception)**
 - **Size of antenna is related to RF frequency of transmission and reception**
 - **Shape (structure) of the antenna is related to RF radiation pattern**
- **Radiation patterns of different antenna types**
- **Positioning antennas**
 - **Maximum coverage of workspace**
 - **Security issues**
- **Real antenna types: omni-directional, semi-directional, highly-directional**

Omnidirectional antenna

- **Omni-directional antenna: radiates RF power equally in all directions around the vertical axis.**
- **Most common example: dipole antenna (see Access Points)**
 - See how to make it (disclaimer: do not try this at home):
<http://www.nodomainname.co.uk/Omnicolinear/2-4collinear.htm>
<http://www.tux.org/~bball/antenna/>
 - Info & fun: <http://www.wlan.org.uk/antenna-page.html>
 - More info: <http://www.hdtvprimer.com/ANTENNAS/types.html>

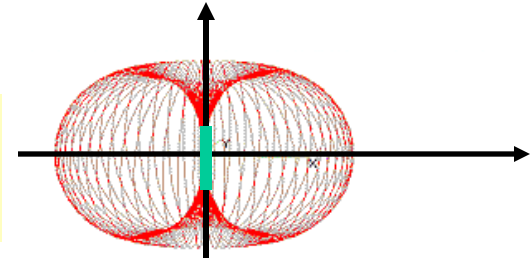


TV dipole

Q: Why TV dipole is bigger?
A: 100 Mhz vs. 2.4 Ghz



AP dipole

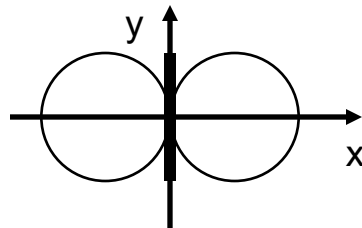


Omnidirectional antennas: simple dipoles

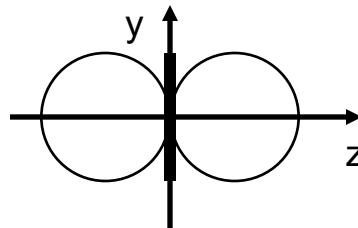
- Real antennas are not isotropic radiators but, e.g., dipoles
→ shape of antenna proportional to wavelength



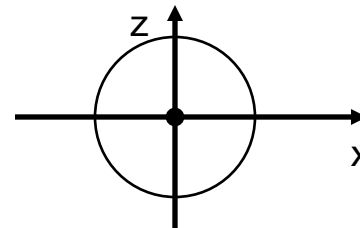
- Example: Radiation pattern of a simple Hertzian dipole



side view (xy-plane)



side view (yz-plane)



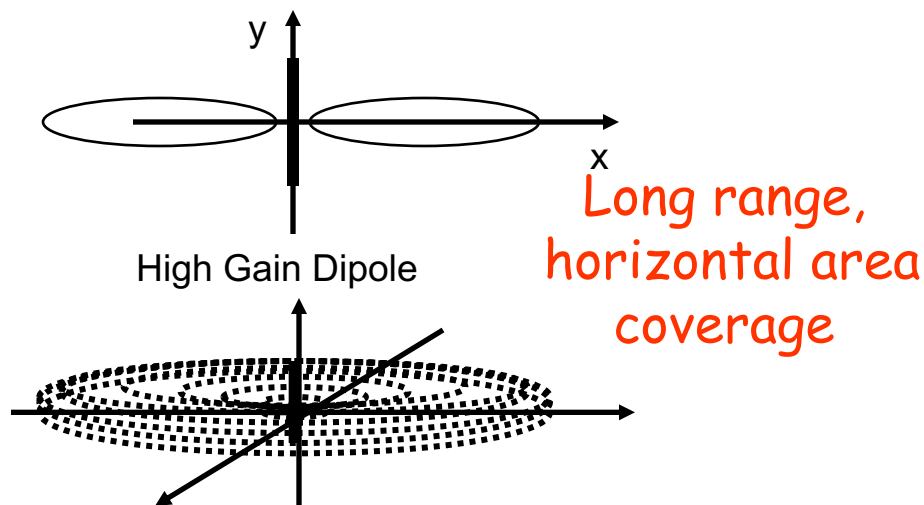
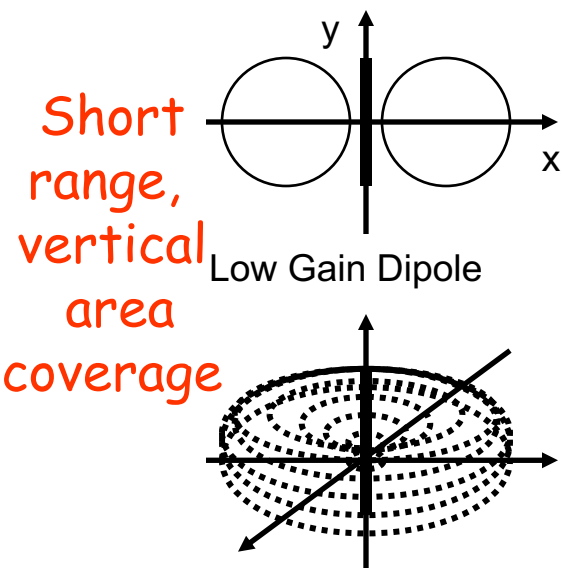
top view (xz-plane)

simple
dipole

- Gain: maximum power in the direction of the main lobe compared to the power of an isotropic radiator (with the same average power)

Omnidirectional antennas: simple dipoles

- Dipole: passive gain is due to concentration (shape) of radiation

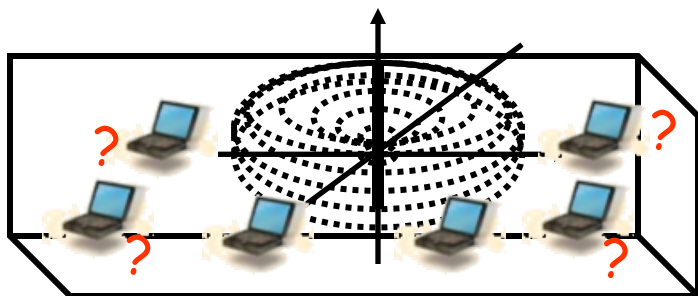


- Dipole: active gain is obtained with power amplifiers (needs external source of energy)
- N.B. near (below) the dipole the signal is weak! And better radiation is obtained in sub-areas around the dipole!

Omnidirectional antennas: simple dipoles

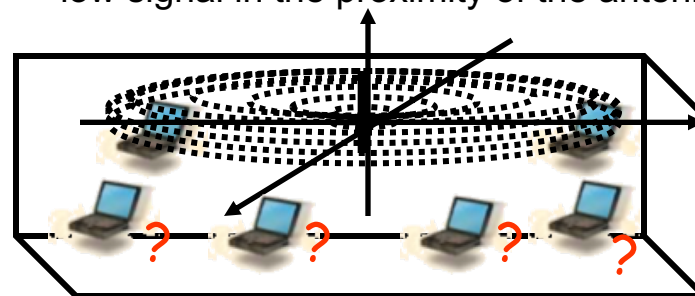
- **Problem: how and when to mount omnidirectional antennas?**
And which gain is ok?

Low Gain Dipole (e.g. 2 dBi)



Room A

High Gain Dipole (e.g. 8..10 dBi), very flat coverage
low signal in the proximity of the antenna



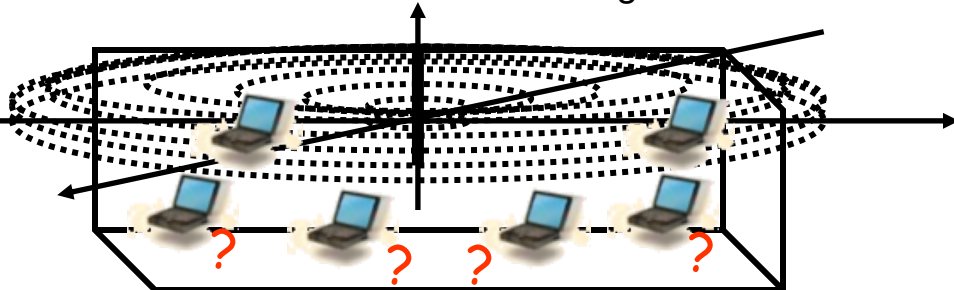
Room B

- **How: Ceiling? Wall? Client positions? Area? Many factors influence the planning...**
- **When:**
 - need for uniform radio coverage around a central point
 - **Outdoor: point-to-multipoint connection (star topology)**

Omnidirectional antennas: simple dipoles

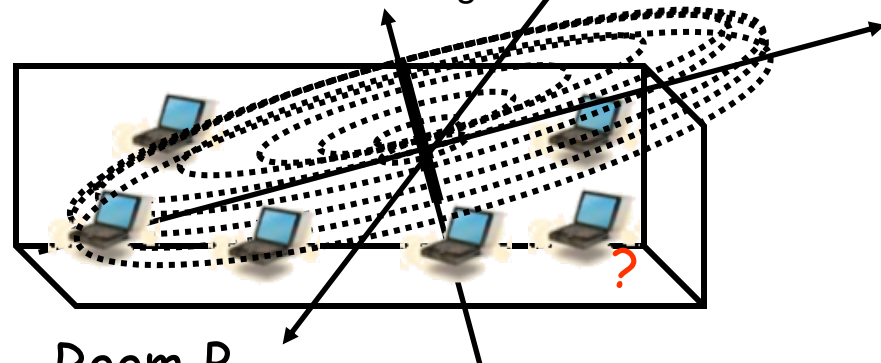
- **Antenna Tilt:** degree of inclination of antenna with respect to perpendicular axis

High Gain Dipole (e.g. 8..10 dBi), very flat coverage mounted on the ceiling



Room A

High Gain Dipole (e.g. 8..10 dBi), very flat coverage mounted on the ceiling with downtilt

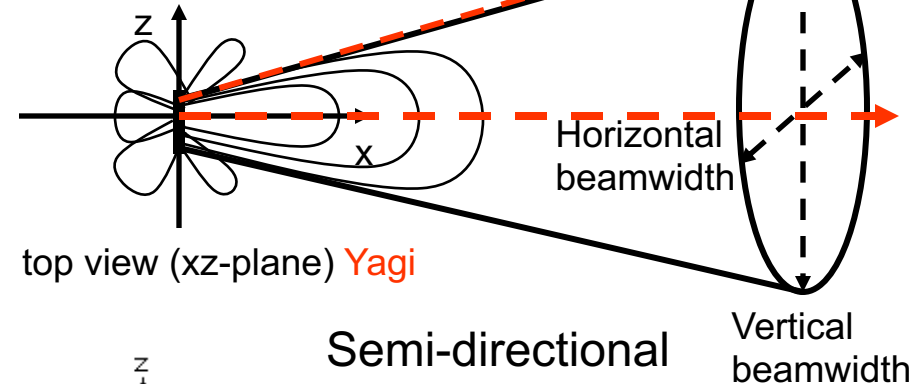
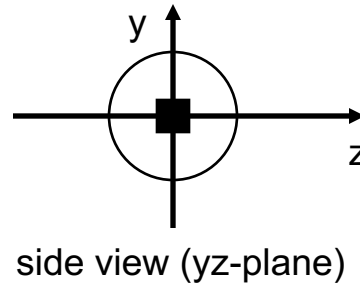
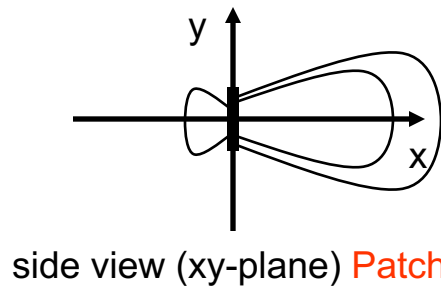


Room B

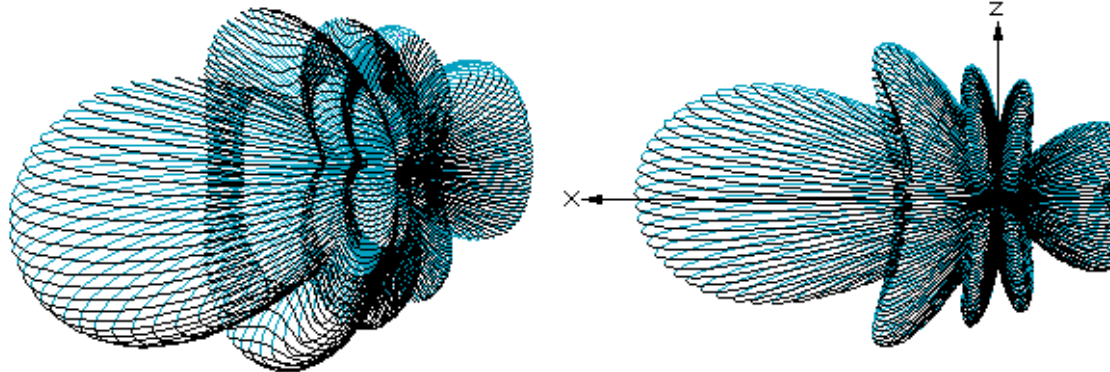
- Some antennas allow a variable degrees **downtilt**.
- Half signal dispersed “in the sky”, 2nd half better exploited.

Semi-directional antennas

- Patch (flat antennas mounted on walls)
- Panel (flat antennas mounted on walls)
- Yagi (rod with tines sticking out)



Semi-directional antenna



Beamwidth cone:
-3dB signal boundary
off-axis

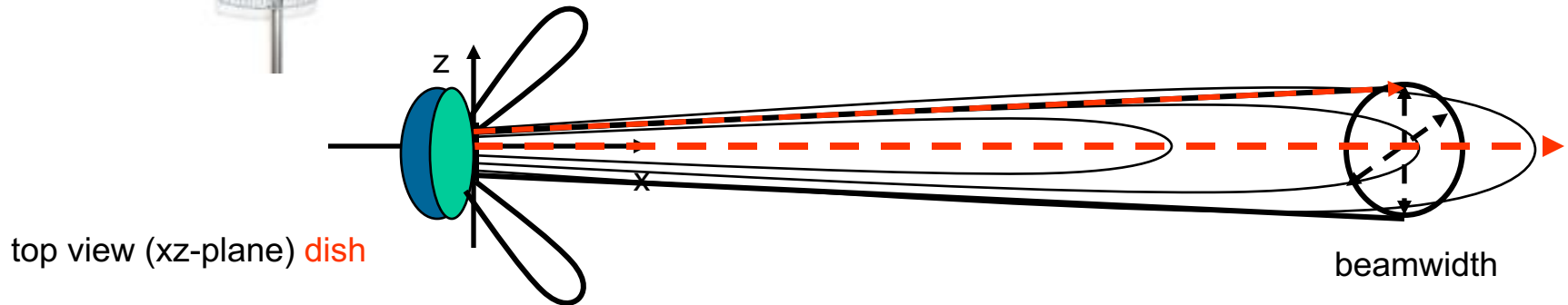
Credits: <http://www.hdtvprimer.com/ANTENNAS/types.html>

highly-directional antennas

- **Parabolic Dish**



- **grid**



Antenna type	H beamwidth	V beamwidth
Omni-dir.	360°	7°.. 80°
Patch/panel	30° .. 180°	6° .. 90°
Yagi	30° .. 78°	14° .. 64°
Parabolic dish	4° .. 25°	4° .. 21°

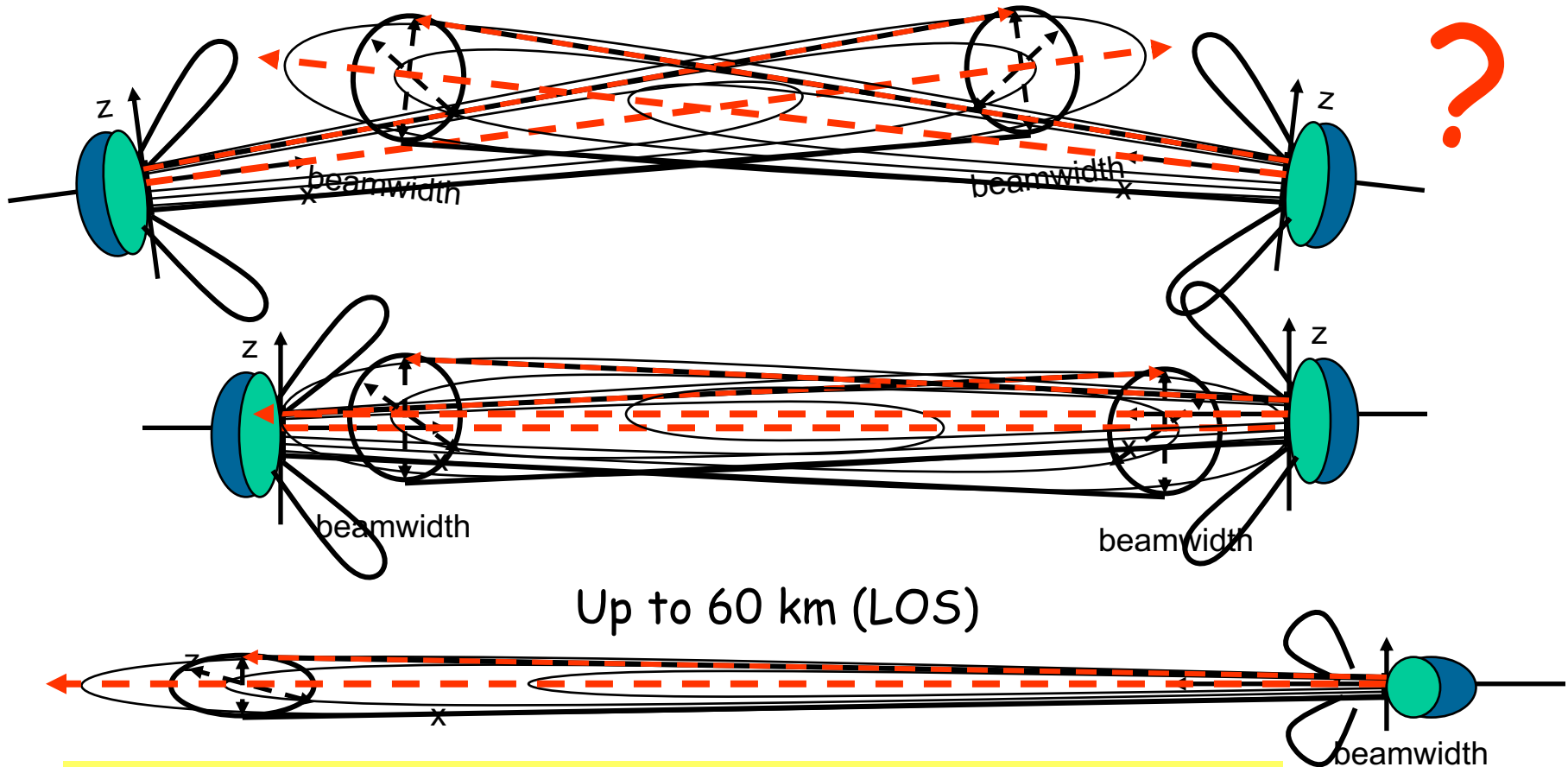
Semi-directional antenna

Beamwidth cone:
-3dB signal boundary
off-axis

highly-directional antennas

- Common use: Point-to-point link

Out of beam alignment

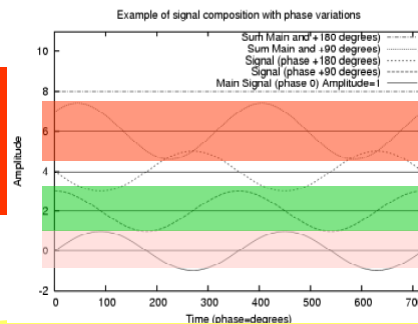


Wind effect: better to have lower gain and wider beam

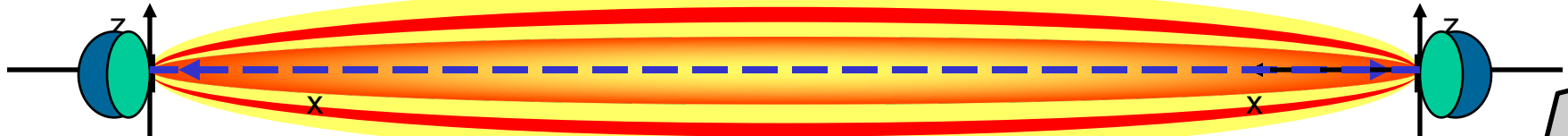
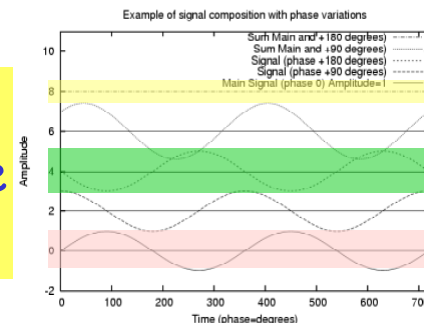
highly-directional antennas

- **Line of sight (LOS):**
 - Straight line between transmitter and receiver
 - No obstructions (outdoor long range reduces reflections)
 - Polarization is more important than in indoor scenarios
- **Fresnel Zone: RF is not laser light, RF signals diffuse energy in space**
 - Ellipse shaped area centered on the LOS axis
 - Most additive RF signal is concentrated in the Fresnel Zone
 - It is important that Fresnel Zone is free from obstacles

Red zone:
additive phase signal



Yellow zone:
inverse phase
signal



highly-directional antennas

■ Fresnel Zone (FZ)

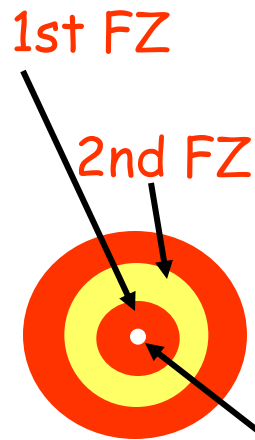
- Blockage of Fresnel Zone causes link disruption
 - Caused by buildings, (growing) trees, foliage, etc.
 - Rule of thumb: < 20% obstruction of Fresnel Zone
 - Practical rule: calculate the radius of FZ leaving 60% unobstructed radius

$$- R_{60\%} = 43.3 \times \sqrt{(d/4f)}$$

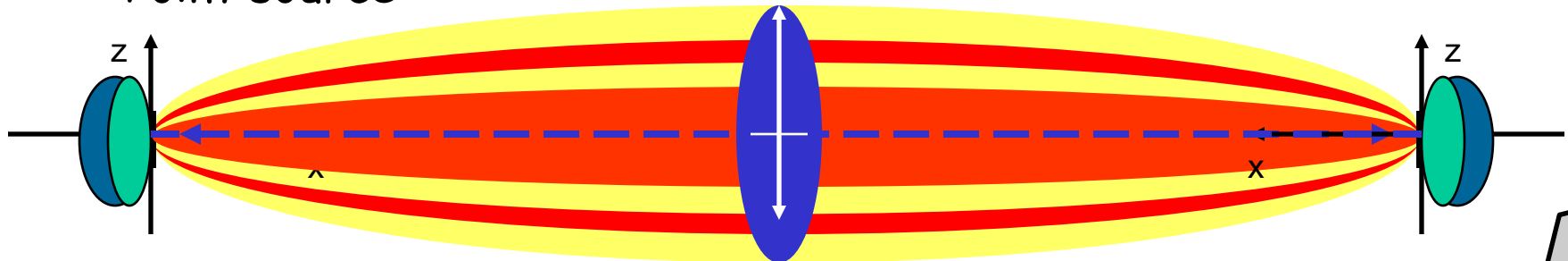
R=radius of 60% central FZ (feet)
d=distance(Miles), f=freq (GhZ)

$$- R_{100\%} = 72.2 \times \sqrt{(d/4f)}$$

R=radius of 100% FZ (feet)
d=distance(Miles), f=freq (GhZ)



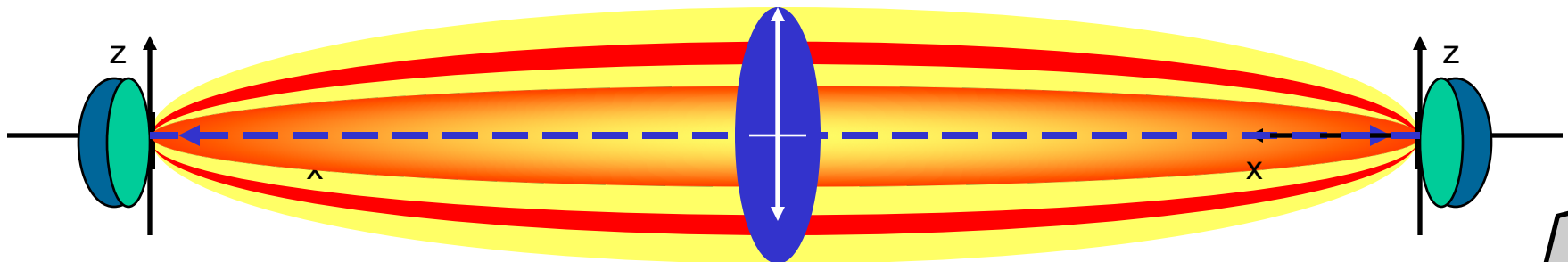
Point source



highly-directional antennas

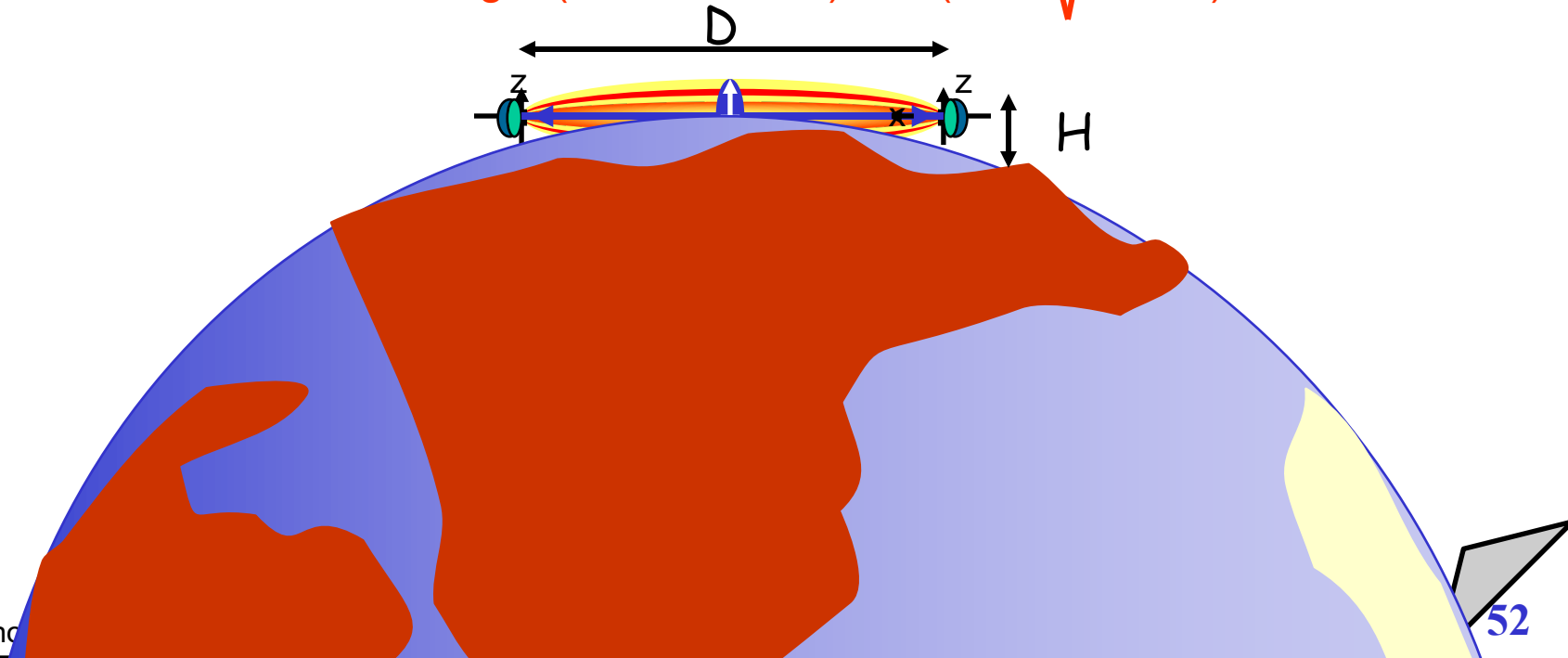
■ Fresnel Zone (FZ)

- N.B. the FZ radius depends only on the distance d between antennas, and frequency f of RF signal!
- Type of antenna, beam width (degree), and gain (dBi) have no effects!
 - E.g. +13 dBi Yagi (30 degree beam) vs. +24 dBi Dish (5 degrees) have the same FZ!!!!
- In practice: if FZ is partially obstructed, it is not useful to use higher gain antennas (with small degree beam) !!!



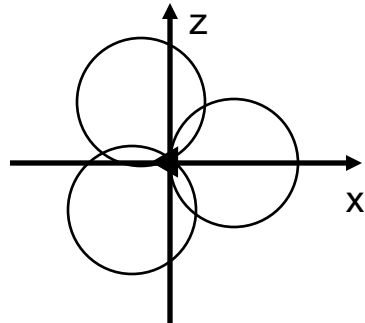
highly-directional antennas

- **Fresnel Zone (FZ)**
 - Is not relevant in indoor scenarios (due to reflections...)
- **Consider the Earth bulge!!!**
 - Very long point-to-point connections may have more than 40% FZ obstructed by Earth surface! Earth Bulge height = h (feet) = $D^2/8$
 - Minimum antenna height (link > 7 miles) $H = (43.3 \sqrt{D/4F}) + D^2/8$

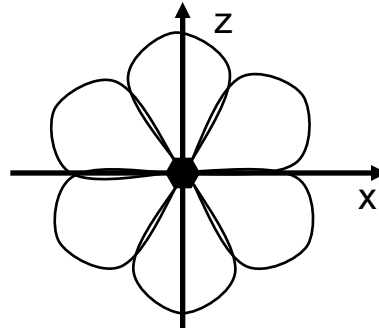


Sectorized-directional antennas

- **Arrays of sectorized directional antennas**



top view, 3 sector



top view, 6 sector

sectorized
antenna

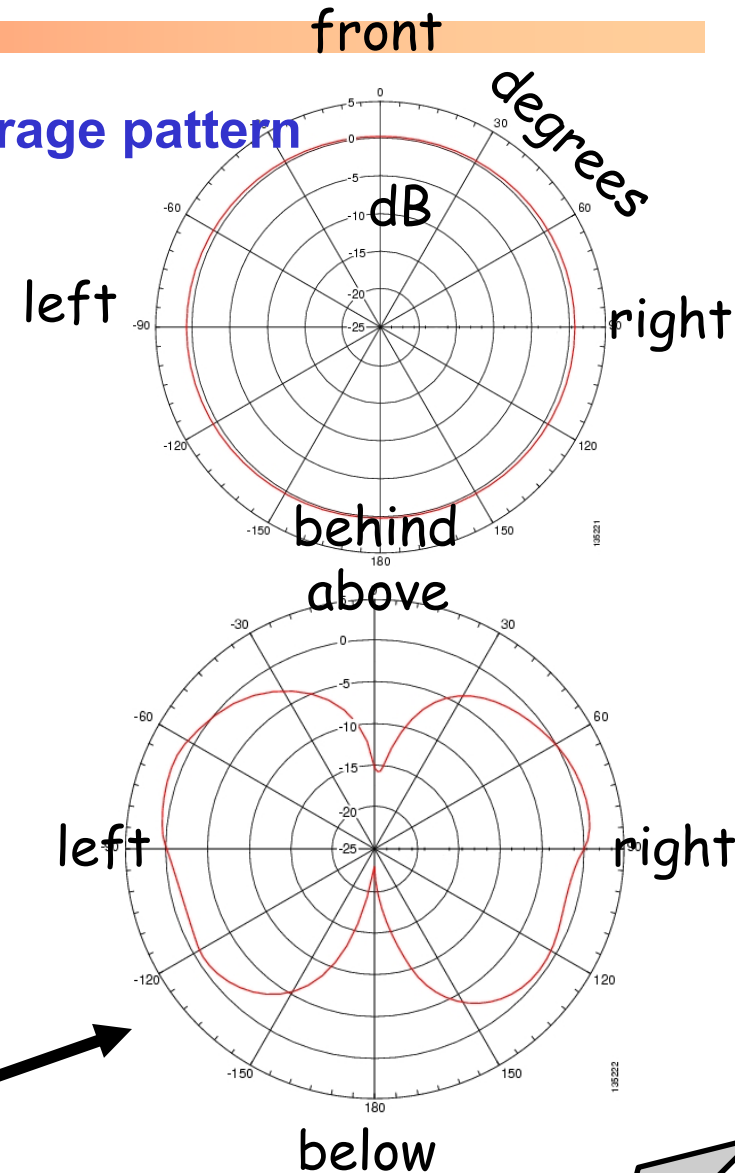


- **Space multiplexing (channel reuse)**

Azimuth and Elevation antenna charts

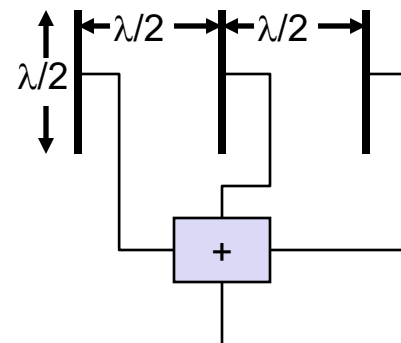
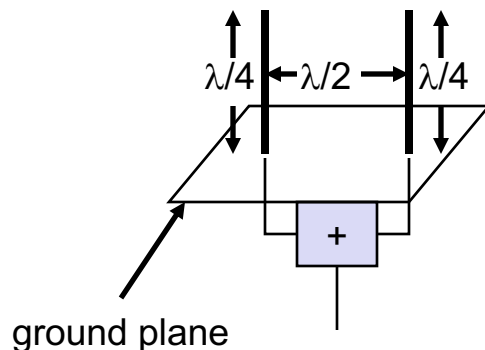
Charts for understanding antenna coverage pattern

- **Azimuth chart** (pattern seen from front/right/behind/left)
 - Obtained with spectrum analyzer with central antenna frequency
 - Signal measured in dB around the antenna
 - E.g. Dipole pattern: almost circular
 - E.g. yagi pattern: high in front, low beside
- N.B. distance and Tx power is not relevant (signal strength in a location is relative to every other location in the chart, like with dB)
- **Elevation chart** (pattern seen front/below/behind/above)



Antennas: diversity

- **Grouping of 2 or more antennas**
 - multi-element antenna arrays
- **Antenna diversity**
 - switched diversity, selection diversity
 - receiver chooses antenna with largest output
 - diversity combining
 - combine output power to produce gain
 - cophasing needed to avoid cancellation (phased antenna array... Requires processor)



Path Loss

- **Path Loss: RF signal “dispersion” (attenuation) as a function of distance**
 - E.g. Possible formulas (**36.6** or **32.4**)
 - Free space: Loss (in dB) = **36.6** + (20 * log₁₀(F)) + (20 * log₁₀(D))
 - F (Mhz), D (miles)
- **Link budget issue: 6 dB rule**
 - Each 6 dB increase in EIRP (signal x 4) implies double Tx range (e.g. see table below: 2.4Ghz Path Loss vs distance)

100 meters	- 80.23 dB	↖ -6 dB
200 meters	- 86.25 dB	
500 meters	- 94.21 dB	↖ -6 dB
1000 meters	- 100.23 dB	
2000 meters	- 106.25 dB	↖ -6 dB
5000 meters	- 114.21 dB	
10000 meters	- 120.23 dB	↖ -6 dB

Link Budget Calculation

- **“Link Budget” or “System Operating Margin”**
 - Excess of signal between transmitter and receiver
 - Calculated for outdoor point-to-point connections
 - Measured in dB (relative) or dBm or mW (absolute)
 - Calculation:
 - Receiver sensitivity RS (weakest detectable signal)
 - The lower the better: e.g. IEEE 802.11 card (see device manual), -95 dBm (1Mbps), -93 dBm (2 Mbps), -90 dBm (5.5 Mbps), -87 dBm (11 Mbps)
 - Link Budget: received power (in dBm) - RS (in dBm)
 - E.g. RS = -82 dBm, received power = -50 dBm
 - Link budget = $-50 - (-82) = +32$ dBm
 - This means the signal has margin of +32 dB before it becomes unviable
- **Fade margin: extra margin for link budget (to cope with multipath variation in indoor/outdoor scenarios): typical [+10..+20] dB**

Link Budget Calculation: example

- Example: design of transmission system, needs amplifier?

